ARBITER: Supplementary material

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Methods appendix

Algorithms and validation

The official 22 August 2019 RoB 2 algorithms were implemented for D1 randomisation, D2 deviations (effect of assignment), D3 missing outcome data, D4 outcome measurement and D5 selection of the reported result. Numbered questions map to the 22 exported signalling questions. Required NA or missing answers make a nonempty path invalid; absent labels or wholly absent answer sets are not evaluable. Overall is the worst domain judgement, with High permitted when at least two domains have Some concerns. The guidance was cross-checked with ARBITER 1.0.0 source (commit 2f549ae). Unit tests cover the decision paths and required-answer checks. Bundle proposals reproduce 510/510 domain outputs; no overrides were recorded. All required active answers in the ARBITER domain histories are present. Unassisted invalid-path sensitivity recursively resolves required invalid answers as N/PN, then NI; off-path answers are not imputed.

Input parity

Both arms received each trial folder as delivered, including the main article, supplements and sources list. For the original 105-trial runs, the Unassisted inputs also included the exact Markdown projection of the registry records shown in the ARBITER run (74 trials), and the 60 protocol and statistical analysis plan PDFs acquired from ClinicalTrials.gov. These inputs were copied from the ARBITER run; nothing was fetched again. For De Waal 2021, the registry record supplied to both arms belongs to a different trial. This shared input error means that registry availability does not establish relevance to the assessed trial. Source: no-kit-sonnet-5.5/README.md.

API list-price reconstruction

Costs are frozen-rate API list-price equivalents, not expenditure. USD rates per million tokens are: Sonnet input 2.00, 1-hour cache write 4.00, 5-minute write 2.50, cache read 0.10, output 10.00; Rates were frozen from the provider pricing documentation on 18 October 2026. Main-thread assistant events are deduplicated by message identity within turn; synthetic zero-usage events are excluded. Thinking tokens are included in output. Side calls are total recorded model usage minus terminal usage. One-hour side-call cache writes are primary; five-minute writes are a sensitivity. Reconstruction bounds coincide. Routing was global, with no geography multiplier. For each trial, equivalent US$ = (2 × input + 4 × 1-hour cache writes + 2.5 × 5-minute cache writes + 0.1 × cache reads + 10 × output tokens) / 1,000,000, summed over recorded model calls. Runs used a subscription; these equivalents are not subscription expenditure. Harness costs are unusable: unrecognised model and unknown cost basis; retained only as supplementary diagnostics.

Error localisation

Losses are cells where both adjudicators agreed where a source label differs from preferred. A reference adjudicator is usable only when their independent judgement equals the adjudicators' agreed choice. Primary requires both; sensitivity at least one. Y/PY collapse to Y and N/PN to N; NI and NA remain separate. Discordance requires all usable adjudicators to agree on a parsed answer different from the losing answer. Adjudicator disagreements are separately flagged, never errors. A single replacement is decisive if it changes a valid algorithm output, even if the new output is not preferred. Other answers, including inactive NA, stay fixed; invalid substituted paths are not decisive. First applicable cell class: judgement departs from own answers; one question alone changed the rating; several questions each changed the rating alone; jointly decisive (all differing replacements reach preferred); no differing questions with complete comparisons; not attributable. The no-discordance class can contain adjudicator-disagreement questions. Reference-judgement departures from their own answers are flagged. Frequencies count all questions that alone changed the rating regardless of cell class; joint sets contribute none. Ambiguous or qualified answers are not imputed. Published losses cannot be localised without signalling answers.

Evidence binding and verification

Each distinct evidence identity cited in bases or counterevidence by current active answers is counted once per bundle; superseded histories are excluded. Visual records (figure kind or render field) take precedence over text spans. Text spans (narrative kind or quote/start/end fields) require a nonblank quote and source version, source identity matching a captured bundle source, finite page/start/end, and start less than end. Quotes are server-extracted spans cited by handle. Visual binding requires render source matching the record and a captured source, positive integer page, nonblank PNG hash, four finite region coordinates in [0,1] with ordered corners, and nonblank transcription. Images are server-rendered and hash-bound; transcriptions are model-written and are not machine-checked against the image. Unknown or missing records are other and unbound in the overall denominator. Empty bundles contribute no cited records. Trials using visual evidence use all 102 trials as denominator. These are structural checks, without PDF re-extraction or a judgement of whether a passage supports an answer. Bundle verification does not establish clinical correctness or completion. Unassisted evidence traces are not available.

Inference and references

All estimates use the fixed 102-trial set. CIs use 10,000 trial-cluster bootstrap replicates stratified by meta-analysis, percentile intervals (quantile type 7), seed 20261018; the same weights support paired differences. Missing predictions count as wrong. For ordinal coefficients a missing prediction receives maximal disagreement; the same rule applies to all estimates. Independent signalling-answer agreement uses the earlier exploratory Y/PY, N/PN and NI/NA collapse. Higgins JPT, Savović J, Page MJ, Sterne JAC, editors. Revised Cochrane risk-of-bias tool for randomized trials (RoB 2), 22 August 2019.
https://www.riskofbias.info/welcome/rob-2-0-tool/current-version-of-rob-2. Salman A et al. ARBITER 1.0.0, public source release, commit 2f549ae.
https://github.com/AliSalman-et-al/ARBITER/tree/2f549ae30f056ff441ea61a04b719968716f2219. Anthropic. Claude API pricing. Accessed 18 October 2026. https://platform.claude.com/docs/en/about-claude/pricing.

Published-label target matching

LABEL-TARGETS.csv links each published label to its review, source table, outcome, time point and contrast, with quoted binding evidence where stated and an explicit inference otherwise. The assessment prompt targets the labelled result rather than automatically the trial primary outcome, with the definition, printed result and locator from the trial report. Among the 102 included trials, 33 targets match an explicitly stated result, 11 match one of several covered outcomes, and 32 match an inferred result; 7 partly match, 9 differ and 10 have unknown binding. These categories are retained rather than treating all targets as exact matches. Twelve published label sets (60 domain labels) cover every outcome taken from a trial; another 10 label sets (50 domain labels) have no identified result, and seven more label sets (35 domain labels) jointly cover pain and disability; thus 29 sets (145 domain labels) do not isolate a single result (19 broad and 10 unknown). Binding was inferred rather than

explicitly stated for 62 label sets, overlapping these categories. Source: benchmark README-FOR-ALI.md, LABEL-TARGETS.csv and per-review BINDING.md; label-target-summary.csv recomputes matching categories.

Dates and training exposure

Trial reports span publication years 1996–2025; the 16 source reviews were published in 2025–2026. Years come from TRIAL-SUMMARY.csv and trial-label years where that field is blank; the benchmark README identifies the SANCTUARY report as 2025. Anthropic documents June 2026 as the training-data cut-off for claude-sonnet-5-5 (model documentation, https://platform.claude.com/docs/en/models/sonnet-5-5/overview, and provider help, https://support.claude.com/en/articles/8114494-how-up-to-date-is-claude-s-training-data; checked 19 October 2026). Dates cannot establish whether a report was used in training.

Adjudication process and primary endpoint

Two blinded adjudicators used no AI. Each rated the domain from the supplied documents and froze the independent rating before seeing any label. They then saw only the three labels, randomised in order for each cell, and chose the preferred label with a reason. They did not see model rationales or quotations. The primary endpoint is the share of adjudicated arm-disagreement cells where the adjudicators preferred ARBITER's rating. Independent-rating paired differences added post hoc compare ARBITER and Unassisted agreement with the same frozen independent rating on all 154 disputed cells, using the shared stratified trial-cluster bootstrap.

Limitations

Adjudication selected cells on disagreement, so preferred shares are subject to discrepant-analysis selection and do not estimate full-benchmark accuracy. Adjudicators chose only among the offered labels; cells where both arms matched Published were not audited. Labels alone may have revealed an arm, although this risk was low because rationales and quotations were hidden. Each arm ran once; workflow, prompt and runtime contributions cannot be separated. Reports or reviews may have appeared in training before the documented June 2026 cut-off. Whether quotations support the answers was not assessed. Overall was not adjudicated. Some label-target bindings were inferred, broad or mismatched; disagreement with a published label may partly reflect outcome mismatch rather than a different judgement.

eTable 1. Benchmark trials span sixteen meta-analyses

Specialty	Journal and year	Topic	Trials	Published D1–D5 labels (Low/SC/High)	Overall given
General medicine	JAMA 2026	SGLT2 inhibitors and kidney outcomes across eGFR and albuminuria	7	35 / 0 / 0	No
General medicine	Lancet 2025	Clopidogrel vs aspirin for secondary prevention of coronary artery disease	7	32 / 3 / 0	Yes
Cardiology	J Am Coll Cardiol 2025	Early aortic valve replacement vs surveillance in asymptomatic severe aortic stenosis	2	10 / 0 / 0	Yes
Endocrinology	Lancet Diabetes Endocrinol 2025	GLP-1 receptor agonists and kidney and cardiovascular outcomes	7	35 / 0 / 0	Yes
Gastroenterology	Gastroenterology 2025	GLP-1 receptor agonists and gastrointestinal adverse events	7	31 / 4 / 0	Yes
Infectious diseases	Clin Infect Dis 2025	Safety of high-dose rifampin in tuberculosis treatment	6	30 / 0 / 0	Yes
Nephrology	Clin J Am Soc Nephrol 2025	Therapies for proteinuria and eGFR decline in IgA nephropathy	7	30 / 5 / 0	No
Neurology	Lancet Neurol 2025	Pharmacotherapy and non-invasive neuromodulation for neuropathic pain	7	24 / 7 / 4	Yes
Obstetrics and gynaecology	Hum Reprod Update 2026	Preconception lifestyle interventions for women	7	23 / 10 / 2	Yes
Oncology	Lancet Oncol 2026	CDK4/6 inhibitors plus endocrine therapy in HR-positive, HER2-negative advanced breast cancer	7	35 / 0 / 0	Yes
Orthopaedics	Am J Sports Med 2025	Lateral extra-articular procedure added to primary ACL reconstruction	5	20 / 3 / 2	Yes
Paediatrics	JAMA Pediatr 2025	Active treatment vs expectant management of patent ductus arteriosus in preterm infants	7	29 / 6 / 0	Yes
Psychiatry	Lancet Psychiatry 2026	Cannabinoids for mental and substance use disorders	5	19 / 6 / 0	Yes
Pulmonary and critical care	Lancet Respir Med 2025	Preoxygenation strategies for intubation in critically ill patients	7	32 / 0 / 3	No
Rheumatology	Lancet Rheumatol 2025	Long-term effectiveness of non-surgical interventions for chronic low back pain	7	12 / 23 / 0	Yes
Surgery and anaesthesia	Br J Anaesth 2025	Home-based prehabilitation before surgery	7	30 / 2 / 3	Yes
Total			102	427 / 69 / 14	81 / 102

102 trials; SC, Some concerns. Overall given indicates availability of published Overall labels. ACL, anterior cruciate ligament; CDK4/6, cyclin-dependent kinases 4 and 6; eGFR, estimated glomerular filtration rate; GLP-1, glucagon-like peptide-1; HER2, human epidermal growth factor receptor 2; HR, hormone receptor; IgA, immunoglobulin A; SGLT2, sodium-glucose cotransporter 2. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 2. Published label distributions by domain

Specialty	Topic	Trials	D1	D2	D3	D4	D5
General medicine	Clopidogrel vs aspirin for secondary prevention of coronary artery disease	7	7/0/0	4/3/0	7/0/0	7/0/0	7/0/0
General medicine	SGLT2 inhibitors and kidney outcomes across eGFR and albuminuria	7	7/0/0	7/0/0	7/0/0	7/0/0	7/0/0
Cardiology	Early aortic valve replacement vs surveillance in asymptomatic severe aortic stenosis	2	2/0/0	2/0/0	2/0/0	2/0/0	2/0/0
Endocrinology	GLP-1 receptor agonists and kidney and cardiovascular outcomes	7	7/0/0	7/0/0	7/0/0	7/0/0	7/0/0
Gastroenterology	GLP-1 receptor agonists and gastrointestinal adverse events	7	6/1/0	6/1/0	7/0/0	7/0/0	5/2/0
Infectious diseases	Safety of high-dose rifampin in tuberculosis treatment	6	6/0/0	6/0/0	6/0/0	6/0/0	6/0/0
Nephrology	Therapies for proteinuria and eGFR decline in IgA nephropathy	7	7/0/0	7/0/0	6/1/0	6/1/0	4/3/0
Neurology	Pharmacotherapy and non-invasive neuromodulation for neuropathic pain	7	6/1/0	6/1/0	3/3/1	7/0/0	2/2/3
Obstetrics and gynaecology	Preconception lifestyle interventions for women	7	3/3/1	7/0/0	5/1/1	6/1/0	2/5/0
Oncology	CDK4/6 inhibitors plus endocrine therapy in HR-positive, HER2-negative advanced breast cancer	7	7/0/0	7/0/0	7/0/0	7/0/0	7/0/0
Orthopaedics	Lateral extra-articular procedure added to primary ACL reconstruction	5	2/1/2	4/1/0	5/0/0	4/1/0	5/0/0
Paediatrics	Active treatment vs expectant management of patent ductus arteriosus in preterm infants	7	6/1/0	3/4/0	7/0/0	6/1/0	7/0/0
Psychiatry	Cannabinoids for mental and substance use disorders	5	4/1/0	3/2/0	3/2/0	4/1/0	5/0/0
Pulmonary and critical care	Preoxygenation strategies for intubation in critically ill patients	7	7/0/0	7/0/0	4/0/3	7/0/0	7/0/0
Rheumatology	Long-term effectiveness of non-surgical interventions for chronic low back pain	7	3/4/0	5/2/0	0/7/0	3/4/0	1/6/0
Surgery and anaesthesia	Home-based prehabilitation before surgery	7	4/1/2	5/1/1	7/0/0	7/0/0	7/0/0

Each D1–D5 cell gives Low / Some concerns / High counts; 102 trials. Source journals and years are in the preceding benchmark eTable. ACL, anterior cruciate ligament; CDK4/6, cyclin-dependent kinases 4 and 6; eGFR, estimated glomerular filtration rate; GLP-1, glucagon-like peptide-1; HER2, human epidermal growth factor receptor 2; HR, hormone receptor; IgA, immunoglobulin A; SGLT2, sodium-glucose cotransporter 2. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 3. AVATAR complete trace

Part A. Answers and judgements

Question	ARBITER	Evidence handles (type)	Justification	Unknowns	Counterevidence	Unassisted
D1 judgement / rationale	Proposed: Low; adopted: Low					Low Allocation used a computer-generated, centre-stratified permuted-block sequence delivered through a web-based interactive response system, so concealment was adequate. The baseline differences in age (P = .02) and haemoglobin (P = .01) are consistent with chance across about 25 variables.
1.1 Random allocation sequence	Yes	eh_d6c269341c7d8a5d (direct support)	The original AVATAR paper states the assignment was computer-generated and stratified by centre using a permuted-block sequence with variable block size, which is a random component.	—	—	Yes
1.2 Concealment until enrolment and assignment	Yes	eh_d6c269341c7d8a5d (direct support)	Patients were randomized using a web-based interactive response system, a central remote allocation method independent of enrolment staff. Variable block size limits predictability of the last allocations in a block.	Whether the central system was run by a unit independent of enrolling staff is not stated explicitly.	—	Yes
1.3 Preintervention imbalance suggesting a randomisation problem	No	eh_92eed84c7b98e37c (direct support); eh_58abdac52262aa33 (context)	Group sizes were 78 vs 79. Of about 25 baseline comparisons in Table 2, only age (P=.02) and haemoglobin (P=.01) were nominally significant, and diabetes was P=.07. The age difference is 1 year in medians. This is compatible with chance and none approaches P<0.001.	—	Basis 1: Diabetes was less common in early surgery (17.9% vs 29.1%), which could favour surgery, but P=.07 is compatible with chance and the 2022 paper attributes it to chance.	Probably no
D2 judgement / rationale	Proposed: Low; adopted: Low					Low The trial was open label, so participants and clinicians knew the assignments. Surgery in the conservative arm followed

Question	ARBITER	Evidence handles (type)	Justification	Unknowns	Counterevidence	Unassisted
						protocol-defined triggers such as symptom onset. Six of 78 early-surgery patients did not receive surgery, and some surgeries were delayed by COVID-19. These are not deviations likely to affect the outcome. The analysis was intention-to-treat.
2.1 Participants aware of assignment	Yes	eh_975aac3202107531 (direct support); eh_3ce4199af9d487f3 (context)	The registry lists masking as NONE, and the comparison is surgical valve replacement versus watchful waiting, which cannot be hidden from participants.	—	—	Yes
2.2 Carers or intervention providers aware of assignment	Yes	eh_975aac3202107531 (direct support)	Open-label surgery versus conservative care means clinicians and surgeons delivering care knew the assignment. Allocation was concealed until assignment, but blinding afterwards was impossible.	—	—	Yes
2.3 Deviations arising from trial context	Probably no	eh_c8e0958840636206 (direct support); eh_22d01d9384e1a677 (context); eh_4b5761bf573cc287 (context); eh_05e667d2d41d3858 (context)	Crossover from conservative care to surgery (35/79) was driven by protocol-defined indications (symptom onset, AS progression, LVEF decline), as Table 3 and the protocol show, so it is consistent with the protocol rather than trial-context driven. Surgical delays beyond 8 weeks arose from the COVID-19 public-health emergency and were documented as an anticipated minor deviation in the SAP amendment. Four early-surgery patients did not receive surgery as assigned, for reasons like infection or declining, which could occur outside a trial.	Reasons for declining surgery are not described in detail.	Basis 3: The authors note pandemic-related follow-up substitution by phone and lengthened AVR delays could have increased clinical risk. These arose from the pandemic, not the trial context, and no deaths occurred while awaiting scheduled surgery in the conservative group.	Probably no
2.4 Deviations affecting the outcome	Not asked (inactive; branching)					Not applicable
2.5 Deviations balanced across groups	Not asked (inactive; branching)					Not applicable
2.6 Appropriate analysis of	Yes	eh_30fa222ec5523dcc (direct support);	All analyses were intention-to-treat as randomized, and all 157 patients were included. One early-surgery	—	—	Yes

Question	ARBITER	Evidence handles (type)	Justification	Unknowns	Counterevidence	Unassisted
assignment effect		eh_c8e0958840636206 (direct support)	patient lost to follow-up was censored and still analysed.			
2.7 Substantial impact of analysing outside assigned groups	Not asked (inactive; branching)					Not applicable
D3 judgement / rationale	Proposed: Low; adopted: Low					Low All 157 patients were analysed. One patient lost to follow-up was censored.
3.1 Outcome data available for nearly everyone	Yes	eh_e66d374e838e8c6d (direct support)	All 157 randomized patients were analysed and median follow-up was 63 months. Only one patient (early-surgery arm) was lost to follow-up and censored at last contact, so 1/157 (0.6%) has incomplete outcome data. This is far fewer than the observed deaths, so any outcome for that patient could not materially change the HR of 0.44.	—	—	Yes
3.2 Evidence against bias from missing data	Not asked (inactive; branching)					Not applicable
3.3 Missingness potentially depends on true outcome	Not asked (inactive; branching)					Not applicable
3.4 Missingness likely depends on true outcome	Not asked (inactive; branching)					Not applicable
D4 judgement / rationale	Proposed: Low; adopted: Low					Low Death is an objective outcome. Assessors were not blinded to allocation, but this cannot plausibly bias an all-cause death count.
4.1 Inappropriate measurement method	Probably no	eh_3ce4199af9d487f3 (direct support)	All-cause death is a pre-specified, objective event ascertained through scheduled follow-up (6-monthly in year 1, then yearly), phone contact when in-person visits were not possible, and review of medical records for any registered event. Nothing suggests an insensitive method or an instrument with poor validity.	—	—	No
4.2 Measurement	Probably no	eh_3ce4199af9d487f3	Both arms followed the same protocol schedule, and	Whether vital	—	No

Question	ARBITER	Evidence handles (type)	Justification	Unknowns	Counterevidence	Unassisted
differs across groups		(direct support)	COVID-related phone follow-up applied to patients regardless of arm. Death is hard to miss, so any difference in contact opportunities (e.g. surgery visits) is unlikely to change ascertainment of death.	status was verified from registries in both arms is not stated.		
4.3 Assessors aware of received intervention	Yes	eh_3ce4199af9d487f3 (direct support)	The trial was open-label (registry masking: none), and the report states that DSMB members were not blinded to allocation when reviewing events. Site investigators recording deaths were also aware of the allocation.	—	—	Not applicable
4.4 Assessment potentially influenced by awareness	Probably no	eh_3ce4199af9d487f3 (inference)	All-cause death is an observer-reported outcome that involves no judgement, so knowledge of the intervention is unlikely to influence its assessment, even though the assessors were aware of the allocation.	—	—	Not applicable
4.5 Assessment likely influenced by awareness	Not asked (inactive; branching)					Not applicable
D5 judgement / rationale	Proposed: Some concerns; adopted: Some concerns					Low Time to death was a prespecified secondary endpoint with a registered 5-year time frame. It was analysed with a single Cox model, and the SAP amendment only addresses COVID-19 delays. I found no sign of selective reporting. A small concern is that the extended follow-up was a later analysis, but it matches the registered time frame.
5.1 Analysis plan finalised before unblinding	No information	eh_d54a2600ddc753b3 (context); eh_7ad155913c1c5826 (context); eh_30fa222ec5523dcc (context); eh_7961ea6680167af2 (context)	The registry entry (first submitted 30 Apr 2015, before enrolment began in June 2015) names all-cause death only as a component of composite outcomes. It does not list time to all-cause death as a separate outcome. The Circulation 2022 and EHJ 2024 reports describe 'time to death' as a prespecified secondary end point analysed with Kaplan-Meier and Cox models, citing a published protocol. That protocol is not among the captured Sources, and the report statements are not date-stamped before unblinded	Whether the published protocol listed time to all-cause death and its Cox analysis, and its date relative to unblinded data.	—	Probably yes

Question	ARBITER	Evidence handles (type)	Justification	Unknowns	Counterevidence	Unassisted
			data. The only SAP-type document is an amendment dated 4 Jan 2021 about COVID surgical delays. No dated source therefore specifies this measurement and analysis.			
5.2 Result selected from eligible outcome measurements	Probably no	eh_30fa222ec5523dcc (direct support); eh_d54a2600ddc753b3 (context)	All-cause death is a single, unambiguous event, so there is essentially one way to measure the outcome domain. The registered time frames were 36 months and 5 years. The 2024 report states that all-cause death was analysed over the full follow-up and also restricted to 3 years, so no eligible time point appears to have been omitted in favour of a more favourable one.	—	—	Probably no
5.3 Result selected from eligible analyses	Probably no	eh_30fa222ec5523dcc (direct support); eh_f625637d366df2a0 (context)	Time-to-event secondary outcomes were analysed with the same Kaplan-Meier, log-rank and Cox model (treatment only) used for the primary outcome, and the same method was used in the 2022 paper. No covariate-adjusted or alternative models are described, and the long-term and 3-year analyses are both reported.	—	—	Probably no
Overall	Proposed: Some concerns; adopted: Some concerns		Any domain Some concerns, with no High domain: Overall Some concerns.			Low

Part B. Evidence records

Handle	Source	Page	Lines	Source-version hash	Full verbatim quote
eh_05e667d2d41d3858	ehae585.pdf	9	76–83	a890f771fd05	follow-up and lengthening AVR delays. Pre-defined physical visits were substituted by phone follow-up. Such surveillance within the watchful waiting may hypothetically impact timely detection of symptoms conse quently leading to increased clinical risk, including sudden death. Nevertheless, in the conservative group, none of the patients experi enced MACE or died while awaiting the scheduled surgery. It is of note that in the small number of patients undergoing TAVR, delays ap peared shorter as compared with the surgical intervention.
eh_22d01d9384e1a677	Current registry-linked sap: SAP_000.pdf	2	1–12	ba0a93294998	Amendmen to statystical analysis 1. Patients randomized to AVR surgery who received delayed surgery (more than 8 weeks after randomization) due to COVID -19 public health emergency will be documented and this delay will be defined as an anticipated minor deviation from the clinical protocol 2. Appropriate statistical

Handle	Source	Page	Lines	Source-version hash	Full verbatim quote
					procedures will be performed and results will be published according to the FDA guidance "Conduct of Clinical Trials of Medical Products During the COVID-19 Public Health Emergency Guidance for Industry, investigators, and Institutional Review Boards"
eh_30fa222ec5523dcc	ehae585.pdf	5	13–24	a890f771fd05	Statistical analysis We conducted all analyses according to the intention-to-treat principle, where all patients were included in the analysis as they were randomized. For the analysis of the primary outcome, we built a Kaplan-Meier graph with a log rank P-value to test for a difference between early surgery and control intervention at a two-sided α value of 0.05 and used a Cox propor tional hazards model to derive hazard ratios (HRs) with 95% confidence in tervals (CIs) for treatment effect estimation. Time-to-event secondary outcomes were analysed using the same meth ods applied for the analysis of the primary outcome. For MACE, all-cause death, HF hospitalization, and cardiovascular death, we repeated the ana lysis as described above restricted to a 3-year follow-up, with complete
eh_3ce4199af9d487f3	ehae585.pdf	3	1–116	a890f771fd05	Introduction Aortic stenosis (AS) is the single most common valvular heart disease re quiring intervention in developed countries and is projected to double by 2050 in both the USA and Europe.1 The approach to symptomatic pa tients with severe AS is straightforward. Surgical aortic valve replacement (AVR) and, more recently, transcatheter aortic valve implantation (TAVI) procedures are Class I recommendation to relieve symptoms and im prove survival in symptomatic patients with severe AS.2,3 Conversely, the question of when and how to treat truly asymptom atic patients with severe AS and normal left ventricular (LV) systolic function is still subject to debate and ongoing research. To date, two randomized trials have demonstrated benefit of early AVR in compari son with conservative strategy in asymptomatic patients with severe and very severe AS patients.4,5 In addition, there are several ongoing randomized trials investigating the role of early TAVI and/or early sur gical AVR in asymptomatic AS. The Aortic Valve Replacement Versus Conservative Treatment in Asymptomatic Severe Aortic Stenosis (AVATAR) trial is an investigator- initiated, prospective, multinational, randomized, controlled, parallel-group, event-driven trial that compared the safety and efficacy of early surgery in the treatment of truly asymptomatic patients with severe AS and LV ejec tion fraction (LVEF) ≥ 50% vs. conservative treatment with watchful wait ing strategy and AVR only after symptom onset, or other guidelines directed Class I or II indications for AVR.3,6 In the primary analysis of the AVATAR trial, asymptomatic AS patients randomized to early surgery had a lower incidence of the composite primary outcome comprising all- cause death, acute myocardial infarction, stroke, or unplanned hospitaliza tion for heart failure (HF), compared with patients who were randomized to conservative treatment.5 We now report the results of the AVATAR trial after extended follow-up at 5 years. Methods Study design and oversight The AVATAR trial (ClinicalTrials.gov identifier NCT02436655) protocol and protocol update were designed by the first co-authors and steering committee co-chairmen and was approved by the steering committee of the AVATAR trial.7,8 The clinical and outcomes data were collected by per sonnel at the participating sites and were sent directly to the data coordinat ing centre at the University Clinical Center of Serbia. No extramural funding was used to support this work. The authors, members of the steering com mittee, and investigators are solely responsible for the design and conduct of this trial, all analyses, drafting and editing of the article, and its final contents. Independent statistician conducted the long-term data analyses. The trial was conducted in accordance with the Declaration of Helsinki. All participants provided written informed consent. An independent data and safety monitoring

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					board (DSMB) adjudicated all serious adverse events and oversaw the safety of the trial. The article was prepared by the first author and steering committee co-chairmen and was reviewed and edited by mem bers of the steering committee and authors. All authors reviewed the article, approved its submission for publication, and vouched for the accuracy and completeness of the data and for the fidelity of the trial to the protocol. Study population The study inclusion and exclusion criteria are shown in Table 1; trial flowchart is shown in Figure 1. Briefly, a total of 157 patients with severe AS according to the guidelines were enrolled in the trial: 79 were randomized to conservative treatment and 78 were randomized to early surgery. To prove truly asymp tomatic status, exercise testing was performed in all candidates to evaluate symptom status according to a standardized protocol using treadmill or semi- supine ergo-bicycle. To consider exercise testing negative, which was manda tory for inclusion, all candidates needed to reach a projected submaximal heart rate. Positive exercise test included onset of AS-related symptoms, fall in systolic blood pressure (≥20 mmHg from the baseline values), or Electrocardiography (ECG) or stress echocardiography signs of myocardial ischaemia.9 Follow-up and trial endpoints Patients were followed according to the protocol every 6 months for the first year and then yearly with the in-person visits at the participating study centre. As the trial was affected by the Covid-19 pandemic, contact with patients who could not come in person was made by phone. For any event that was registered, the medical records were asked for and reviewed. Adverse clinical events were adjudicated by the DSMB per protocol defini tions.7,8 Data and safety monitoring board members were not blinded to the treatment allocation during events review. They adjudicated the events by consensus. Table 1 Eligibility inclusion/exclusion criteria Inclusion criteria • Men and women of any ethnic origin aged ≥18 years • Written informed consent • Vmax across the aortic valve > 4 m/s or P mean ≥ 40 mmHg and AVA ≤ 1 cm2 or AVAi ≤ 0.6 cm2/m2 at rest • Without reported symptoms • Society of Thoracic Surgeons score < 8% Exclusion criteria Participation in another clinical trial within 30 days prior to randomization • Pregnant or nursing women • Mental condition rendering the patient unable to understand the nature, scope, and possible consequences of the study or to follow the protocol • Positive stress-test defined as follows: (a) Anginal chest pain during testing (b) Syncope or dizziness during testing (c) Decrease in systolic blood pressure during exercise ≥ 20 mmHg (d) Malignant arrhythmia during exercise testing (VT or VF) • Left ventricular ejection fraction < 50% at rest • Very severe AS defined as Vmax > 5.5 m/s at rest • Significant disease of other valves: mitral stenosis with Pmean > 5 mg, or any significant regurgitation ≥ 3+ • Recent previous myocardial infarction (<1 year) • Need for additional aortic root replacement (i.e. Bentall) or ascending aorta surgery in asymptomatic patients undergoing AVR • Previous coronary bypass surgery • Previous any heart valve surgery • Impaired renal function, i.e. creatinine > 200 µmol/L or glomerular filtration rate < 30 mL/min/1.73 m2 • Significant pulmonary hypertension at rest (systolic pulmonary artery pressure > 50 mmHg) • Uncontrolled hypertension at rest (systolic blood pressure > 180 mmHg and diastolic blood pressure > 100 mmHg) • Significant comorbidity with reduced life expectancy (<3 years) • Uncontrolled diabetes mellitus (HbA1c > 9%) • Significant chronic obstructive pulmonary disease (FEV1 < 70% of predicted value)
eh_4b5761bf573cc287	CIRCULATIONAHA.121.057639.pdf	4	8–15	592b9a6c531c	Patients in the conservative treatment group were referred for surgery in case of onset of AS-related symptoms, if the LVEF decreased to <50%, or if the peak aortic jet velocity increased each year by >0.3 m/s on follow-up echocardiog- raphy, according to European Society of Cardiology guide- lines on valvular

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					disease at the time the study was designed.11 Conservative treatment included the treatment of risk factors and comorbidities.
eh_58abdac52262aa33	ehae585.pdf	5	47–50	a890f771fd05	Diabetes mellitus 14 (17.9%) 23 (29.1%) .07
eh_7961ea6680167af2	Current registry-linked sap: SAP_000.pdf	1	1–8	ba0a93294998	Avatar trial protocol -Amendment to surgical delays- Aortic Valve Replacement Versus Conservative Treatment in Asymptomatic Severe Aortic Stenosis (AVATAR trial) NCT number : NCT02436655. Date of amendment: 04.01.2021
eh_7ad155913c1c5826	CIRCULATIONAHA.121.057639.pdf	4	16–35	592b9a6c531c	Trial End Points The primary end point was a composite of all-cause mortality or major adverse cardiovascular events (MACEs) composed of acute myocardial infarction, stroke, and unplanned HF hospital- ization needing intravenous treatment with diuretics or inotro- pes. Prespecified secondary end points included the following: • In-hospital and 30-day postoperative mortality in oper- ated patients in both groups • Repeat aortic valve surgery in operated patients in both groups • Repeated MACEs, including stroke, acute myocardial infarction, or unplanned HF hospitalization needing IV diuretic treatment • Major bleeding defined as types 3, 4, and 5 according to the consensus report from the Bleeding Academic Research Consortium13 • Thromboembolic complications on the basis of clinical symptoms, signs, and imaging studies • Time to death • Time to first HF hospitalization
eh_92eed84c7b98e37c	ehae585.pdf	5	26–136	a890f771fd05	Table 2 Selected baseline characteristics of the study population Early surgery group (n = 78) Conservative treatment group (n = 79) P-value Demographics and comorbidities Age, years, median (IQR) 68 (63-73) 69 (64-75) .02 Sex (female) 32 (41.0%) 35 (44.3%) .67 STS score, median (IQR) 1.6 (1.1-2.2) 1.8 (1.2-2.7) .67 Days from randomization to surgery, median (IQR) 55 (36-79) 476 (226-1098) <.001 Diabetes mellitus 14 (17.9%) 23 (29.1%) .07 Hypertension 69 (88.4%) 70 (88.6%) .44 History of PCI 1 (1.3%) 2 (2.5%) .44 History of stroke 2 (2.5%) 2 (2.5%) .92 Peripheral arterial disease 0 (0%) 1 (1.36%) .80 Laboratory parameters BNP, pg/mLa, median (IQR) 83 (53-127) 89 (58-149) .61 NT-proBNP, pg/mLa, median (IQR) 381 (153-660) 347 (186-722) .45 Haemoglobin, g/L, median (IQR) 141 (131-150) 134 (128-141) .01 Creatinine, μmol/L, median (IQR) 80 (66-94) 76 (67-92) .27 Blood glucose, mmol/L, median (IQR) 5.6 (5.3-6.7) 5.6 (5.1-6.8) .70 Echocardiography LVESV, mL, median (IQR) 28 (20-41) 33 (22-42) .96 LVEDV, mL, median (IQR) 113 (89-142) 113 (96-126) .54 LV mass index, g/m2, median (IQR) 152 (133-175) 160 (139-183) .67 Left atrium, cm 4.1 (3.8-4.4) 4.2 (3.9-4.4) .68 SVi, mL/m2, median (IQR) 39 (33-48) 42 (34-51) .58 PA systolic pressure, mmHg, median (IQR) 30 (26-36) 30 (27-37) .82 Vmax, m/s, median (IQR) 4.5 (4.3-4.8) 4.5 (4.2-4.7) .13 Pmean, mmHg, median (IQR) 51 (44-58) 50 (43-59) .16 Pmax, mmHg, median (IQR) 82 (74-90) 79 (70-90) .18 AVA, cm2, median (IQR) 0.73 (0.55-0.84) 0.74 (0.59-0.89) .29 AVAi, cm2/m2, median (IQR) 0.37 (0.3-0.42) 0.37 (0.31-0.46) .08 E/e', median (IQR) 12.2 (9.6-16.3) 12.2 (8.8-18.1) .54
eh_975aac3202107531	ClinicalTrials.gov NCT02436655	1	67–68	d32e23eb2a1e	#### Masking info - Masking: NONE
eh_af1340e6c28a70eb	ehae585.pdf	7	1–148	a890f771fd05	group was 67.9 years, while the mean age of patients in the early surgery group was 65.4 years (P = .19). Four patients in the conservative treat ment group underwent TAVI procedure. Indications for surgery in the conservative treatment group are shown in Table 3. Median time to sur gery upon establishing the indication was 123 days (IQR 90-297 days). There were no significant baseline differences between the patients in the

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					conservative group who remained AVR-free during follow-up and those who later underwent AVR (see Supplementary data online, Table S1). Patients in the conservative group on the watchful waiting, in cluding those undergoing valve surgery, experienced a progressive increase in the aortic valve severity associated with a significant decline in LVEF over time (see Supplementary data online, Tables S2 and S3, respectively). In total, 57/107 (53.3%) patients received a bioprosthetic valve and 50/107 received a mechanical valve (46.7%). In the early surgery group, 34/72 (47.2%) received a bioprosthetic valve, while 23/35 (65.7%) pa tients received a bioprosthetic valve in the conservative treatment group (P = .32). Three patients in early surgery group (3/78; 3.8%) and four pa tients in the conservative treatment group (4/35; 11.4%) have undergone concomitant coronary artery bypass grafting. No peri-procedural myo cardial infarction has been noticed. The intraoperative mortality was low in both groups (1.3% in the early surgery vs. 2.5% in the conservative group; P = .54). There was no difference in length of hospitalization in the early surgery group vs. operated patients in the conservative treatment group (8.8 vs. 9.1 days; P = .87). Details of post-operative aortic valve function, procedural time, and mean hospitalization length between both groups are shown in Supplementary data online, Table S4. Three pa tients (3/78, 3.8%) in the early surgery group, and 1/35 (2.86%) patient in the conservative treatment group underwent redo AVR procedure (P = .79; Supplementary data online, Table S5. Outcomes There were a total of 55 primary outcome events comprising a com posite of all-cause mortality or acute myocardial infarction, stroke, and unplanned HF hospitalization. A primary outcome event occurred in 18/78 patients (23.1%) in the early surgery group and in 37/79 pa tients (46.8%) in the conservative treatment group (HR with early sur gery vs. conservative treatment 0.42; 95% CI 0.24-0.73, P = .002; Figure 2). Forty patients died during follow-up: 13/78 (16.7%) in the early surgery group and 27/79 (34.17%) in the conservative treatment group. In the early surgery group, 11/72 patients died after valve surgery at median of 362 days (IQR 116-1211), of which 8 were cardiovascular deaths. In the conservative group, 8/35 patients died after surgery at shorter median of 131 days (IQR 47.8-603), of which 5 were cardiovas cular deaths. Mortality causes for individual subjects are detailed in Supplementary data online, Tables S6A and S6B. Kaplan-Meier estimates of the individual endpoints of all-cause death and HF hospitalization were significantly lower in the early surgery compared with the conservative group (HR 0.44; 95% CI 0.23-0.85, P = .012, for all-cause death and HR 0.21; 95% CI 0.06-0.73, P = .007, for HF hospitalizations) (Figure 3 and Table 4). The risk of repeated MACE was significantly higher in the conservative treatment group (Table 4). Sudden cardiac death occurred in 9/79 patients (11.4%) in the con servative group compared with 4/78 patients (5.1%) in the early surgery group (P = .17; Table 4). The annual rate of sudden cardiac death was twice higher in the conservatively treated group as compared with early surgery group (2.2% vs. 1%; P = .32). If not considering symptomatic pa tients in the watchful waiting group who died suddenly at home, the an nual rate of sudden cardiac death in the conservative group reached 1.48%. One patient randomized to the early surgery group died sud denly while awaiting the surgery. There was a trend for higher rates of cardiovascular deaths in the conservative treatment group over time (19.3% vs. 11.6%; P = .1; Figure 4). The incidence of serious adverse events was also significantly higher in the conservative treatment group (49.4% vs. 26.4%; P = .013) (Table 4). No difference between groups was observed in the safety outcomes including major bleeding and thromboembolic complications (5.1% vs. 3.8%; P = .69; and 5.1% vs. 3.8%; P = .69; respectively; Table 4).

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					In a post hoc heterogeneity analysis in the study population, no signifi cant interaction for heterogeneity was noted for any of the analysed parameters (see Supplementary data online, Figure S1) indicating that results are consistent across different patients' subgroups. Discussion The main findings from the extended follow-up of the AVATAR trial can be summarized as follows: (i) the primary composite outcome comprising of all-cause death, acute myocardial infarction, stroke, and HF hospitalizations was significantly and consistently lower with the early surgery approach compared with the conservative treatment ap proach; (ii) patients on conservative strategy appeared at almost 50% p = 0.01 HR = 0.44 (95% CI 0.23 to 0.85) 0% 10% 20% 30% 40% 50% 60% All-cause death incidence 79 71 61 39 14 Control 78 72 71 52 15 Early surgery Participants, n 0 20 40 60 80 Time (Months) Early surgery Conservative treatment p = 0.007 HR = 0.21 (95% CI 0.06 to 0.73) 0% 10% 20% 30% 40% 50% 60% HF hospitalization incidence 79 65 55 32 11 Control 78 71 70 51 15 Early surgery Participants, n 0 20 40 60 80 Time (Months) Early surgery Conservative treatment Figure 3 Cumulative incidence of all-cause death (upper panel) and heart failure hospitalization (lower panel) (intention-to-treat population) 4532 Banovic et al. Downloaded from academic.oup.com/eurheartj/article/45/42/4526/7742145 by guest on 03 October 2026
eh_c8e0958840636206	ehae585.pdf	6	1–99	a890f771fd05	3-year follow-up information available for all patients. For binary outcomes, we used a generalized linear model assuming a binomial distribution with a log link function to report risk ratios and 95% CI. We conducted subgroup analyses accompanied by a P for interaction to assess the consistency of the treatment effect on the primary outcome according to the following pre- defined variables: age (<65 vs. ≥65), sex, baseline ejection fraction (<60% vs. ≥60%), and country of trial site. P-values and 95% CI for secondary out comes and secondary analyses of the primary outcome were not adjusted for multiple testing and are considered exploratory. For the post hoc ana lyses of echocardiography readouts of valve severity and LVEF in patients in the conservative group, we calculated the mean change from baseline to 12-month follow-up for all patients in the conservative group with avail able data and the mean change from baseline to final follow-up for all pa tients in the conservative group that underwent surgery and had available data. We conducted paired t-tests to derive P-values and 95% CI for the mean changes from baseline.11 This analysis was repeated for patients in the conservative group that underwent AVR. Analyses were performed using Stata, version 18.0. Results Patients and follow-up Main patient characteristics at baseline are summarized in Table 2. The average age of enrolled patients was 67 years, 57% were men, the ma jority of patients had a degenerative aetiology of AS (84.7%), and the median estimated operative mortality according to the Society of Thoracic Surgeons predicted risk of mortality score was 1.7%. No sig nificant heterogeneity between both groups was noted. All patients were included in the analysis. Median follow-up for all patients was 63 months [interquartile range (IQR) 48-75). Median follow-up in sur vivors was 68 months (IQR 58-79). One patient, who was randomized to early surgery, has been lost to follow-up, and this patient was cen sored when the last follow-up information was available and included in the intention-to-treat analysis (Figure 1). Aortic valve replacement procedures In the early surgery group, AVR was performed in 72 of the 78 patients (92.3%). Among the six patients that did not undergo AVR per random ization, one patient, as above mentioned, was lost to follow-up, and one died prior to valve surgery. Four patients were available for the extended follow-up. Among them, one patient died unoperated after 2113 days; two became symptomatic and underwent intervention after 691 and 791 days, respectively; and one remained

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					asymptomatic and AVR free throughout the entire follow-up (Figure 1). The median time from ran domization to AVR in the early surgery group was 55 days (IQR 36-79). Thirty-five out of the 79 patients (44.3%) in the conservative treatment group had surgery during follow-up. Median time from randomization to surgery in the conservative treatment group was 476 days (IQR 226-1098). At the time of AVR, the mean age of patients in the control p = 0.002 HR = 0.42 (95% CI 0.24 to 0.73) 0% 10% 20% 30% 40% 50% 60% Primary composite endpoint incidence 79 61 52 29 11 Control 78 69 67 48 13 Early surgery Participants, n 0 20 40 60 80 Time (Months) Early surgery Conservative treatment Figure 2 Cumulative incidence of primary composite outcome (intention-to-treat population) Table 3 Indications for aortic valve replacement in the conservative treatment group Symptom onset (including increased fatigue) 18 (51.4%) AS progression 6 (17.1%) Decrease in LVEF 3 (8.6%) Combination of factors 8 (22.9%) AS, aortic stenosis; LVEF, left ventricular ejection fraction. Aortic valve replacement vs. conservative treatment 4531 Downloaded from academic.oup.com/eurheartj/article/45/42/4526/7742145 by guest on 03 October 2026 [Extracted table 1] | Symptom onset (including increased fatigue) 1|8 (51.4%)| |---|---| |AS progression 6 (17.1%)|| |Decrease in LVEF|3 (8.6%)| |Combination of factors 8 (22.9%)||
eh_d54a2600ddc753b3	ClinicalTrials.gov NCT02436655	1	95–101	d32e23eb2a1e	### Primary outcome 1: all cause death, Major Adverse Cardiac Event (MACE) including: (Acute Myocardial Infarction - AMI, Stroke - CVI, unplanned hospitalization for Heart failure (HF) needing intravenous treatment - Measure: all cause death, Major Adverse Cardiac Event (MACE) including: (Acute Myocardial Infarction - AMI, Stroke - CVI, unplanned hospitalization for Heart failure (HF) needing intravenous treatment - Time frame: 36 months ### Primary outcome 2: all cause death, Major Adverse Cardiac Event (MACE) including: (Acute Myocardial Infarction - AMI, Stroke - CVI, unplanned hospitalization for Heart failure (HF) needing intravenous treatment - Measure: all cause death, Major Adverse Cardiac Event (MACE) including: (Acute Myocardial Infarction - AMI, Stroke - CVI, unplanned hospitalization for Heart failure (HF) needing intravenous treatment - Time frame: 5 years
eh_d6c269341c7d8a5d	CIRCULATIONAHA.121.057639.pdf	3	46–50	592b9a6c531c	randomly assigned to early surgery or conservative treatment using a web-based interactive response system. The assign- ment to each treatment group was computer-generated and stratified according to the participating centers by means of a permuted-block sequence with variable block size. Details
eh_e66d374e838e8c6d	ehae585.pdf	6	26–32	a890f771fd05	nificant heterogeneity between both groups was noted. All patients were included in the analysis. Median follow-up for all patients was 63 months [interquartile range (IQR) 48-75). Median follow-up in sur vivors was 68 months (IQR 58-79). One patient, who was randomized to early surgery, has been lost to follow-up, and this patient was cen sored when the last follow-up information was available and included in the intention-to-treat analysis (Figure 1).
eh_f625637d366df2a0	CIRCULATIONAHA.121.057639.pdf	4	62–69	592b9a6c531c	were compared by using 2-sample t tests. A Kaplan-Meier estimator was used to estimate the distribution of time to pri- mary and time-to-event secondary end points (eg, survival and time to first HF hospitalization) and a log-rank test to compare them between the 2 treatment groups. A Cox proportional- hazards regression model that included treatment was used to estimate the hazard ratio comparing the early surgery and the conservative treatment groups. For dichotomous second-

Illustrative all-cause-death assessment (Banovic 2024). Part A gives all questions, answers, evidence types, justifications, unknowns, counterevidence and Unassisted rationales. — means none recorded in those fields; inactive questions follow branching. Part B preserves 16 archived records (15 cited by current answers). Quotes and rationales are verbatim; quote line breaks display as spaces. Source

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locations and hashes identify captured versions. References within quotations concern original documents. Full trace: analysis/data-derived/avatar-trace.json. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 4. Confusion matrices against Published

ARBITER (Sonnet 5.5) — D1

Published	Low	Some concerns	High
Low	75	6	3
Some concerns	7	6	0
High	1	3	1

ARBITER (Sonnet 5.5) — D2

Published	Low	Some concerns	High
Low	69	14	3
Some concerns	12	2	1
High	1	0	0

ARBITER (Sonnet 5.5) — D3

Published	Low	Some concerns	High
Low	66	13	4
Some concerns	1	7	6
High	1	2	2

Each grid contains counts of cells: Published rows × predicted columns, ordered Low / Some concerns / High. Missing predictions appear only when nonzero. Overall uses the 81 labelled trials. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 4 (continued). Confusion matrices against Published

ARBITER (Sonnet 5.5) — D4

Published	Low	Some concerns	High
Low	87	6	0
Some concerns	2	6	1
High	0	0	0

ARBITER (Sonnet 5.5) — D5

Published	Low	Some concerns	High
Low	50	31	0
Some concerns	3	15	0
High	0	3	0

ARBITER (Sonnet 5.5) — Overall

Published	Low	Some concerns	High
Low	23	15	4
Some concerns	3	15	4
High	0	8	9

Each grid contains counts of cells: Published rows × predicted columns, ordered Low / Some concerns / High. Missing predictions appear only when nonzero. Overall uses the 81 labelled trials. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 4 (continued). Confusion matrices against Published

Unassisted (Sonnet 5.5) — D1

Published	Low	Some concerns	High
Low	75	7	2
Some concerns	6	7	0
High	1	4	0

Unassisted (Sonnet 5.5) — D2

Published	Low	Some concerns	High
Low	71	11	4
Some concerns	12	2	1
High	0	1	0

Unassisted (Sonnet 5.5) — D3

Published	Low	Some concerns	High
Low	70	7	6
Some concerns	4	6	4
High	4	0	1

Each grid contains counts of cells: Published rows × predicted columns, ordered Low / Some concerns / High. Missing predictions appear only when nonzero. Overall uses the 81 labelled trials. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 4 (continued). Confusion matrices against Published

Unassisted (Sonnet 5.5) — D4

Published	Low	Some concerns	High
Low	89	4	0
Some concerns	4	2	3
High	0	0	0

Unassisted (Sonnet 5.5) — D5

Published	Low	Some concerns	High
Low	61	19	1
Some concerns	8	10	0
High	2	0	1

Unassisted (Sonnet 5.5) — Overall

Published	Low	Some concerns	High
Low	32	7	3
Some concerns	12	7	3
High	2	6	9

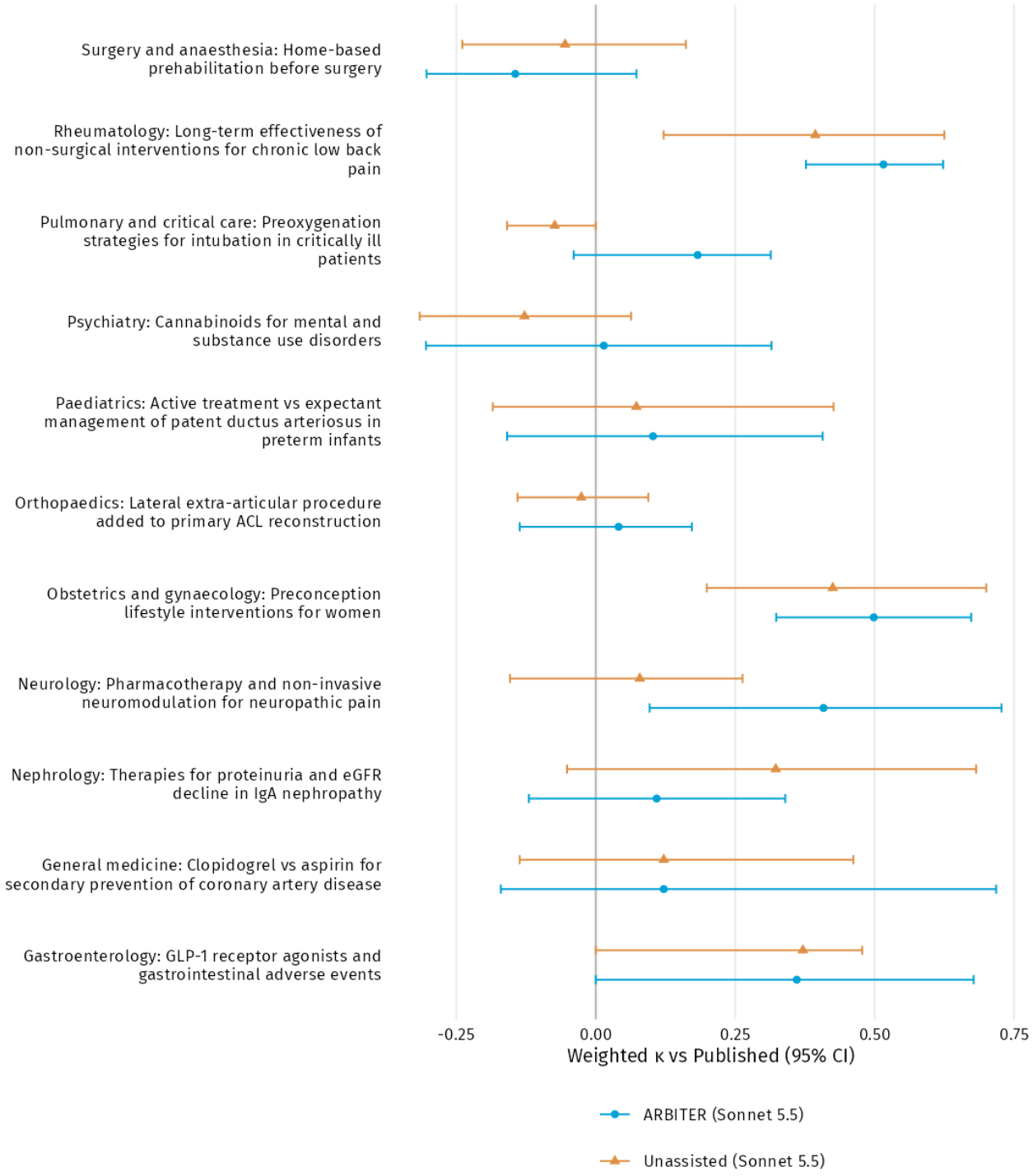
Each grid contains counts of cells: Published rows × predicted columns, ordered Low / Some concerns / High. Missing predictions appear only when nonzero. Overall uses the 81 labelled trials. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 5. Per-meta-analysis weighted κ

Specialty	Topic	ARBITER (Sonnet 5.5)	Unassisted (Sonnet 5.5)
General medicine	Clopidogrel vs aspirin for secondary prevention of coronary artery disease	0.12 (−0.17 to 0.72)	0.12 (−0.14 to 0.46)
General medicine	SGLT2 inhibitors and kidney outcomes across eGFR and albuminuria	—	—
Cardiology	Early aortic valve replacement vs surveillance in asymptomatic severe aortic stenosis	—	—
Endocrinology	GLP-1 receptor agonists and kidney and cardiovascular outcomes	—	—
Gastroenterology	GLP-1 receptor agonists and gastrointestinal adverse events	0.36 (0.00–0.68)	0.37 (0.00–0.48)
Infectious diseases	Safety of high-dose rifampin in tuberculosis treatment	—	—
Nephrology	Therapies for proteinuria and eGFR decline in IgA nephropathy	0.11 (−0.12 to 0.34)	0.32 (−0.05 to 0.68)
Neurology	Pharmacotherapy and non-invasive neuromodulation for neuropathic pain	0.41 (0.10–0.73)	0.08 (−0.15 to 0.26)
Obstetrics and gynaecology	Preconception lifestyle interventions for women	0.50 (0.32–0.67)	0.42 (0.20–0.70)
Oncology	CDK4/6 inhibitors plus endocrine therapy in HR-positive, HER2-negative advanced breast cancer	—	—
Orthopaedics	Lateral extra-articular procedure added to primary ACL reconstruction	0.04 (−0.14 to 0.17)	−0.03 (−0.14 to 0.09)
Paediatrics	Active treatment vs expectant management of patent ductus arteriosus in preterm infants	0.10 (−0.16 to 0.41)	0.07 (−0.18 to 0.43)
Psychiatry	Cannabinoids for mental and substance use disorders	0.01 (−0.30 to 0.32)	−0.13 (−0.32 to 0.06)
Pulmonary and critical care	Preoxygenation strategies for intubation in critically ill patients	0.18 (−0.04 to 0.31)	−0.07 (−0.16 to 0.00)
Rheumatology	Long-term effectiveness of non-surgical interventions for chronic low back pain	0.52 (0.38–0.62)	0.39 (0.12–0.62)
Surgery and anaesthesia	Home-based prehabilitation before surgery	−0.14 (−0.30 to 0.07)	−0.06 (−0.24 to 0.16)

ACL, anterior cruciate ligament; CDK4/6, cyclin-dependent kinases 4 and 6; eGFR, estimated glomerular filtration rate; GLP-1, glucagon-like peptide-1; HER2, human epidermal growth factor receptor 2; HR, hormone receptor; IgA, immunoglobulin A; SGLT2, sodium-glucose cotransporter 2. Weighted κ (95% CI). eFigure 1 displays the defined nondegenerate estimates. A coefficient with no variance, or a degenerate bootstrap interval, is shown as —. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eFigure 1. Agreement with Published varied across meta-analyses



Weighted kappa and 95% CIs; source symbols distinguish model and assistance. Undefined coefficients are omitted. Specialty and topic identify each meta-analysis. Descriptive estimates only; no statistical tests. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018.

eTable 6. Agreement against Published, all sources and domains

ARBITER (Sonnet 5.5)

Domain	Weighted κ	AC2	Exact agreement	Sensitivity	Specificity
D1–D5	0.34 (0.26–0.42)	0.81 (0.78–0.84)	386/510 (75.7%; 72.4–79.0)	55/83 (66.3%; 57.3–74.7)	347/427 (81.3%; 77.7–84.7)
D1	0.38 (0.19–0.55)	0.84 (0.78–0.89)	82/102 (80.4%; 74.5–86.3)	10/18 (55.6%; 35.0–77.8)	75/84 (89.3%; 83.2–95.1)
D2	−0.01 (−0.15 to 0.14)	0.77 (0.69–0.84)	71/102 (69.6%; 61.8–77.5)	3/16 (18.8%; 0.0–37.5)	69/86 (80.2%; 72.6–87.4)
D3	0.43 (0.29–0.57)	0.76 (0.68–0.84)	75/102 (73.5%; 66.7–80.4)	17/19 (89.5%; 73.9–100.0)	66/83 (79.5%; 72.6–86.4)
D4	0.57 (0.33–0.78)	0.95 (0.91–0.98)	93/102 (91.2%; 86.3–96.1)	7/9 (77.8%; 50.0–100.0)	87/93 (93.5%; 88.9–97.8)
D5	0.30 (0.20–0.41)	0.71 (0.64–0.77)	65/102 (63.7%; 56.9–71.6)	18/21 (85.7%; 75.0–95.5)	50/81 (61.7%; 53.1–70.4)
Overall	0.44 (0.32–0.56)	0.50 (0.39–0.60)	47/81 (58.0%; 49.4–66.7)	36/39 (92.3%; 83.7–100.0)	23/42 (54.8%; 44.2–67.4)

Linearly weighted Cohen's κ and Gwet's AC2. Exact agreement, sensitivity and specificity are n/N (%); their CI bounds are percentages. Some concerns or High is the positive class. All-Low rule has no variance; only accuracy and specificity are reported. Paired differences are on the coefficient scale or in percentage points; — indicates undefined or not applicable. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 6 (continued). Agreement against Published, all sources and domains

Unassisted (Sonnet 5.5)

Domain	Weighted κ	AC2	Exact agreement	Sensitivity	Specificity
D1–D5	0.28 (0.21–0.35)	0.82 (0.79–0.85)	395/510 (77.5%; 74.1–80.8)	42/83 (50.6%; 40.3–60.2)	366/427 (85.7%; 82.1–89.3)
D1	0.39 (0.21–0.55)	0.85 (0.79–0.90)	82/102 (80.4%; 74.5–86.3)	11/18 (61.1%; 40.0–81.8)	75/84 (89.3%; 83.3–95.1)
D2	0.06 (−0.10 to 0.21)	0.79 (0.72–0.85)	73/102 (71.6%; 64.7–78.4)	4/16 (25.0%; 6.7–45.5)	71/86 (82.6%; 76.1–89.0)
D3	0.29 (0.13–0.44)	0.76 (0.67–0.83)	77/102 (75.5%; 68.6–82.4)	11/19 (57.9%; 38.9–76.5)	70/83 (84.3%; 77.9–90.7)
D4	0.43 (0.22–0.61)	0.94 (0.90–0.97)	91/102 (89.2%; 83.3–94.1)	5/9 (55.6%; 28.6–85.7)	89/93 (95.7%; 92.0–98.9)
D5	0.25 (0.07–0.43)	0.76 (0.69–0.83)	72/102 (70.6%; 63.7–78.4)	11/21 (52.4%; 34.8–71.4)	61/81 (75.3%; 67.9–82.9)
Overall	0.42 (0.27–0.56)	0.53 (0.43–0.64)	48/81 (59.3%; 50.6–67.9)	25/39 (64.1%; 50.0–77.8)	32/42 (76.2%; 65.9–86.8)

Linearly weighted Cohen's κ and Gwet's AC2. Exact agreement, sensitivity and specificity are n/N (%); their CI bounds are percentages. Some concerns or High is the positive class. All-Low rule has no variance; only accuracy and specificity are reported. Paired differences are on the coefficient scale or in percentage points; — indicates undefined or not applicable. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 6 (continued). Agreement against Published, all sources and domains

All-Low rule

Domain	Weighted κ	AC2	Exact agreement	Sensitivity	Specificity
D1–D5	—	—	427/510 (83.7%; 81.0–86.5)	—	427/427 (100.0%)
D1	—	—	84/102 (82.4%; 76.5–88.2)	—	84/84 (100.0%)
D2	—	—	86/102 (84.3%; 78.4–90.2)	—	86/86 (100.0%)
D3	—	—	83/102 (81.4%; 76.5–86.3)	—	83/83 (100.0%)
D4	—	—	93/102 (91.2%; 86.3–96.1)	—	93/93 (100.0%)
D5	—	—	81/102 (79.4%; 74.5–84.3)	—	81/81 (100.0%)
Overall	—	—	42/81 (51.9%; 44.4–59.3)	—	42/42 (100.0%)

Linearly weighted Cohen's κ and Gwet's AC2. Exact agreement, sensitivity and specificity are n/N (%); their CI bounds are percentages. Some concerns or High is the positive class. All-Low rule has no variance; only accuracy and specificity are reported. Paired differences are on the coefficient scale or in percentage points; — indicates undefined or not applicable. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 7. Paired differences against Published

Domain	Weighted κ	AC2	Exact agreement	Sensitivity	Specificity
ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5)					
D1–D5	0.06 (−0.03 to 0.14)	−0.01 (−0.04 to 0.02)	−1.8 (−5.1 to 1.6)	15.7 (6.8–24.4)	−4.4 (−7.6 to −1.2)
D1	−0.01 (−0.12 to 0.10)	−0.01 (−0.04 to 0.03)	0.0 (−3.9 to 3.9)	−5.6 (−16.7 to 0.0)	0.0 (−2.6 to 2.6)
D2	−0.07 (−0.24 to 0.11)	−0.01 (−0.09 to 0.06)	−2.0 (−9.8 to 5.9)	−6.2 (−26.7 to 13.3)	−2.3 (−11.5 to 6.6)
D3	0.14 (0.01–0.28)	0.00 (−0.05 to 0.06)	−2.0 (−7.8 to 3.9)	31.6 (13.6–47.6)	−4.8 (−10.5 to 0.0)
D4	0.13 (−0.08 to 0.36)	0.01 (−0.02 to 0.05)	2.0 (−3.9 to 7.8)	22.2 (0.0–50.0)	−2.2 (−6.6 to 2.2)
D5	0.05 (−0.15 to 0.24)	−0.05 (−0.13 to 0.02)	−6.9 (−15.7 to 2.0)	33.3 (11.8–52.9)	−13.6 (−22.0 to −5.0)
Overall	0.02 (−0.11 to 0.16)	−0.04 (−0.16 to 0.09)	−1.2 (−11.1 to 8.6)	28.2 (16.1–40.5)	−21.4 (−32.7 to −9.8)

ARBITER minus Unassisted; coefficients in original units, proportions in percentage points. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 8. Post hoc: unweighted agreement with published labels is modest

Source	Domain	Measure	Estimate (95% CI)
ARBITER (Sonnet 5.5)	D1	Unweighted κ	0.37 (0.18–0.53)
ARBITER (Sonnet 5.5)	D1	Exact agreement	82/102 (80.4%; 74.5–86.3)
ARBITER (Sonnet 5.5)	D2	Unweighted κ	−0.02 (−0.16 to 0.14)
ARBITER (Sonnet 5.5)	D2	Exact agreement	71/102 (69.6%; 61.8–77.5)
ARBITER (Sonnet 5.5)	D3	Unweighted κ	0.37 (0.23–0.51)
ARBITER (Sonnet 5.5)	D3	Exact agreement	75/102 (73.5%; 66.7–80.4)
ARBITER (Sonnet 5.5)	D4	Unweighted κ	0.55 (0.30–0.77)
ARBITER (Sonnet 5.5)	D4	Exact agreement	93/102 (91.2%; 86.3–96.1)
ARBITER (Sonnet 5.5)	D5	Unweighted κ	0.28 (0.18–0.39)
ARBITER (Sonnet 5.5)	D5	Exact agreement	65/102 (63.7%; 56.9–71.6)
ARBITER (Sonnet 5.5)	Overall	Unweighted κ	0.37 (0.24–0.49)
ARBITER (Sonnet 5.5)	Overall	Exact agreement	47/81 (58.0%; 49.4–66.7)
Unassisted (Sonnet 5.5)	D1	Unweighted κ	0.38 (0.20–0.54)
Unassisted (Sonnet 5.5)	D1	Exact agreement	82/102 (80.4%; 74.5–86.3)
Unassisted (Sonnet 5.5)	D2	Unweighted κ	0.03 (−0.11 to 0.19)
Unassisted (Sonnet 5.5)	D2	Exact agreement	73/102 (71.6%; 64.7–78.4)
Unassisted (Sonnet 5.5)	D3	Unweighted κ	0.31 (0.15–0.46)
Unassisted (Sonnet 5.5)	D3	Exact agreement	77/102 (75.5%; 68.6–82.4)
Unassisted (Sonnet 5.5)	D4	Unweighted κ	0.34 (0.13–0.55)
Unassisted (Sonnet 5.5)	D4	Exact agreement	91/102 (89.2%; 83.3–94.1)
Unassisted (Sonnet 5.5)	D5	Unweighted κ	0.26 (0.09–0.42)
Unassisted (Sonnet 5.5)	D5	Exact agreement	72/102 (70.6%; 63.7–78.4)
Unassisted (Sonnet 5.5)	Overall	Unweighted κ	0.32 (0.17–0.47)
Unassisted (Sonnet 5.5)	Overall	Exact agreement	48/81 (59.3%; 50.6–67.9)

Post hoc. Unweighted Cohen's κ and exact agreement against published review labels, by domain and Overall. κ here is unweighted; exact agreement uses labelled trials. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 9. Adjudicators usually agreed with models when they disagreed with Published

Source	Direction	Outcome	Count	Consensus n	All n
ARBITER	More lenient	Other label	0	28	36
ARBITER	More lenient	Published upheld	6	28	36
ARBITER	More lenient	Split	8	28	36
ARBITER	More lenient	Adjudicators agreed with it	22	28	36
ARBITER	Stricter	Other label	2	77	88
ARBITER	Stricter	Published upheld	2	77	88
ARBITER	Stricter	Split	11	77	88
ARBITER	Stricter	Adjudicators agreed with it	73	77	88
Unassisted	More lenient	Other label	2	38	46
Unassisted	More lenient	Published upheld	17	38	46
Unassisted	More lenient	Split	8	38	46
Unassisted	More lenient	Adjudicators agreed with it	19	38	46
Unassisted	Stricter	Other label	1	59	69
Unassisted	Stricter	Published upheld	10	59	69
Unassisted	Stricter	Split	10	59	69
Unassisted	Stricter	Adjudicators agreed with it	48	59	69

Both sources use the same model. Figure 4B uses disagreements where both adjudicators agreed as the denominator; cells where adjudicators split are displayed separately. Adjudicators agreed with it means their agreed choice equals the source label; Published upheld means their agreed choice equals the published review label; Other label means neither. Count is the outcome count; Agreed n excludes divided decisions and All n includes them. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 10. Published labels were usually less severe than the agreed choice

Domain	Direction	Eligible cells (n)	Total
D1	More lenient	9	18
D1	Equal	2	18
D1	Stricter	7	18
D2	More lenient	11	32
D2	Equal	11	32
D2	Stricter	10	32
D3	More lenient	19	24
D3	Equal	5	24
D3	Stricter	0	24
D4	More lenient	9	14
D4	Equal	3	14
D4	Stricter	2	14
D5	More lenient	30	41
D5	Equal	8	41
D5	Stricter	3	41
All	More lenient	78	129
All	Equal	29	129
All	Stricter	22	129

Published review label relative to the adjudicators' agreed choice, on the 129 ratings where both adjudicators agreed. Directions compare severity on Low < Some concerns < High. Selected disputed ratings cannot estimate full-benchmark accuracy. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 11. Agreement against the composite reference, Sonnet only

Domain	Weighted κ	AC2	Exact agreement
ARBITER (Sonnet 5.5)			
D1–D5	0.94 (0.91–0.97)	0.98 (0.97–0.99)	472/485 (97.3%; 95.9–98.6)
D1	0.97 (0.92–1.00)	0.99 (0.98–1.00)	97/98 (99.0%; 96.9–100.0)
D2	0.80 (0.61–0.93)	0.96 (0.93–0.99)	88/94 (93.6%; 88.7–97.9)
D3	0.97 (0.92–1.00)	0.98 (0.96–1.00)	92/94 (97.9%; 94.7–100.0)
D4	0.92 (0.80–1.00)	0.99 (0.97–1.00)	99/101 (98.0%; 95.1–100.0)
D5	0.96 (0.91–1.00)	0.98 (0.96–1.00)	96/98 (98.0%; 95.8–100.0)
Post hoc split: Adjudicators' agreed choice	—	—	116/129 (89.9%)
Post hoc split: Kept published label	—	—	356/356 (100.0%)
Unassisted (Sonnet 5.5)			
D1–D5	0.69 (0.62–0.76)	0.91 (0.88–0.93)	425/485 (87.6%; 85.2–90.0)
D1	0.89 (0.77–1.00)	0.97 (0.94–1.00)	95/98 (96.9%; 93.8–100.0)
D2	0.42 (0.14–0.65)	0.88 (0.82–0.92)	78/94 (83.0%; 76.7–89.1)
D3	0.78 (0.65–0.89)	0.90 (0.85–0.95)	83/94 (88.3%; 82.3–93.8)
D4	0.69 (0.48–0.86)	0.96 (0.93–0.98)	94/101 (93.1%; 88.2–97.0)
D5	0.55 (0.44–0.67)	0.80 (0.74–0.86)	75/98 (76.5%; 69.7–83.2)
Post hoc split: Adjudicators' agreed choice	—	—	69/129 (53.5%)
Post hoc split: Kept published label	—	—	356/356 (100.0%)
All-Low rule			
D1–D5	—	—	358/485 (73.8%; 70.6–77.0)
D1	—	—	78/98 (79.6%; 73.5–85.7)
D2	—	—	79/94 (84.0%; 77.7–90.4)
D3	—	—	66/94 (70.2%; 63.5–76.8)
D4	—	—	87/101 (86.1%; 80.4–91.2)
D5	—	—	48/98 (49.0%; 41.7–56.6)

Weighted κ, AC2 and exact agreement. The composite uses the agreed choice on 129 cells and Published on 356; 25 cells where adjudicators split are excluded. Adjudicators chose among three labels; unreviewed cells matching Published are assumed correct. Both inflate absolute values. Post hoc component rows give n/N (%) without new intervals. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 12. Paired differences against the composite reference

Domain	Weighted κ	AC2	Exact agreement
ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5)			
D1–D5	0.25 (0.18–0.33)	0.07 (0.05–0.10)	9.7 (7.0–12.4)
D1	0.08 (0.00–0.20)	0.02 (0.00–0.05)	2.0 (0.0–5.1)
D2	0.38 (0.12–0.68)	0.08 (0.03–0.14)	10.6 (3.2–18.1)
D3	0.18 (0.07–0.32)	0.08 (0.03–0.14)	9.6 (3.4–15.8)
D4	0.23 (0.00–0.48)	0.03 (0.00–0.06)	5.0 (0.0–10.0)
D5	0.41 (0.28–0.52)	0.18 (0.12–0.24)	21.4 (14.3–28.3)

ARBITER minus Unassisted; coefficients in original units, proportions in percentage points. Reference: the adjudicators' agreed choice where both agreed; published labels elsewhere. Excludes 25 divided decisions (485 cells per arm). Only three labels were offered. Unreviewed cells where both arms matched published labels are assumed correct. Both features inflate absolute values. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 13. Composite-reference sensitivity estimates by adjudicator

Adjudicator	Source	Domain	Weighted κ	AC2	Exact agreement
Adjudicator 1	ARBITER (Sonnet 5.5)	D1–D5	0.86 (0.81–0.90)	0.95 (0.93–0.97)	479/510 (93.9%; 92.0–95.9)
Adjudicator 1	ARBITER (Sonnet 5.5)	D1	0.90 (0.81–0.98)	0.97 (0.95–0.99)	98/102 (96.1%; 93.1–99.0)
Adjudicator 1	ARBITER (Sonnet 5.5)	D2	0.70 (0.50–0.85)	0.92 (0.87–0.96)	92/102 (90.2%; 85.3–95.1)
Adjudicator 1	ARBITER (Sonnet 5.5)	D3	0.83 (0.72–0.92)	0.90 (0.84–0.95)	92/102 (90.2%; 85.3–95.1)
Adjudicator 1	ARBITER (Sonnet 5.5)	D4	0.92 (0.80–1.00)	0.99 (0.97–1.00)	100/102 (98.0%; 95.1–100.0)
Adjudicator 1	ARBITER (Sonnet 5.5)	D5	0.90 (0.82–0.96)	0.96 (0.92–0.98)	97/102 (95.1%; 91.2–98.0)
Adjudicator 1	Unassisted (Sonnet 5.5)	D1–D5	0.65 (0.58–0.72)	0.88 (0.85–0.90)	435/510 (85.3%; 82.5–87.8)
Adjudicator 1	Unassisted (Sonnet 5.5)	D1	0.87 (0.75–0.98)	0.97 (0.93–0.99)	98/102 (96.1%; 92.2–99.0)
Adjudicator 1	Unassisted (Sonnet 5.5)	D2	0.35 (0.12–0.55)	0.82 (0.76–0.88)	79/102 (77.5%; 70.6–84.3)
Adjudicator 1	Unassisted (Sonnet 5.5)	D3	0.71 (0.57–0.83)	0.85 (0.78–0.92)	87/102 (85.3%; 79.4–91.2)
Adjudicator 1	Unassisted (Sonnet 5.5)	D4	0.68 (0.50–0.84)	0.95 (0.92–0.98)	94/102 (92.2%; 87.3–96.1)
Adjudicator 1	Unassisted (Sonnet 5.5)	D5	0.54 (0.42–0.66)	0.79 (0.73–0.85)	77/102 (75.5%; 68.6–82.4)
Adjudicator 1	All-Low rule	D1–D5	—	—	364/510 (71.4%; 67.8–74.9)
Adjudicator 1	All-Low rule	D1	—	—	79/102 (77.5%; 70.6–84.3)
Adjudicator 1	All-Low rule	D2	—	—	81/102 (79.4%; 72.5–86.3)
Adjudicator 1	All-Low rule	D3	—	—	69/102 (67.6%; 60.8–74.5)
Adjudicator 1	All-Low rule	D4	—	—	87/102 (85.3%; 79.4–91.2)
Adjudicator 1	All-Low rule	D5	—	—	48/102 (47.1%; 40.2–54.9)
Adjudicator 2	ARBITER (Sonnet 5.5)	D1–D5	0.91 (0.88–0.94)	0.97 (0.96–0.98)	488/510 (95.7%; 94.1–97.1)
Adjudicator 2	ARBITER (Sonnet 5.5)	D1	0.95 (0.87–1.00)	0.99 (0.97–1.00)	100/102 (98.0%; 95.1–100.0)
Adjudicator 2	ARBITER (Sonnet 5.5)	D2	0.71 (0.52–0.85)	0.93 (0.88–0.96)	91/102 (89.2%; 83.3–94.1)
Adjudicator 2	ARBITER (Sonnet 5.5)	D3	0.96 (0.90–1.00)	0.97 (0.95–1.00)	99/102 (97.1%; 94.1–100.0)
Adjudicator 2	ARBITER (Sonnet 5.5)	D4	0.89 (0.78–1.00)	0.98 (0.96–1.00)	99/102 (97.1%; 94.1–100.0)

Adjudicator	Source	Domain	Weighted κ	AC2	Exact agreement
Adjudicator 2	ARBITER (Sonnet 5.5)	D5	0.94 (0.88–1.00)	0.97 (0.95–1.00)	99/102 (97.1%; 94.1–100.0)
Adjudicator 2	Unassisted (Sonnet 5.5)	D1–D5	0.67 (0.59–0.73)	0.89 (0.87–0.91)	439/510 (86.1%; 83.3–88.6)
Adjudicator 2	Unassisted (Sonnet 5.5)	D1	0.82 (0.68–0.94)	0.95 (0.91–0.99)	96/102 (94.1%; 89.2–98.0)
Adjudicator 2	Unassisted (Sonnet 5.5)	D2	0.50 (0.25–0.70)	0.87 (0.82–0.92)	85/102 (83.3%; 77.5–89.2)
Adjudicator 2	Unassisted (Sonnet 5.5)	D3	0.72 (0.58–0.83)	0.86 (0.80–0.92)	86/102 (84.3%; 78.4–90.2)
Adjudicator 2	Unassisted (Sonnet 5.5)	D4	0.73 (0.55–0.88)	0.96 (0.93–0.98)	95/102 (93.1%; 88.2–97.1)
Adjudicator 2	Unassisted (Sonnet 5.5)	D5	0.52 (0.40–0.65)	0.79 (0.73–0.85)	77/102 (75.5%; 68.6–82.4)
Adjudicator 2	All-Low rule	D1–D5	—	—	372/510 (72.9%; 69.6–76.1)
Adjudicator 2	All-Low rule	D1	—	—	81/102 (79.4%; 73.5–85.3)
Adjudicator 2	All-Low rule	D2	—	—	84/102 (82.4%; 76.5–88.2)
Adjudicator 2	All-Low rule	D3	—	—	68/102 (66.7%; 60.8–72.5)
Adjudicator 2	All-Low rule	D4	—	—	87/102 (85.3%; 79.4–91.2)
Adjudicator 2	All-Low rule	D5	—	—	52/102 (51.0%; 44.1–58.8)

510 cells per Sonnet arm. Each adjudicator's selected label replaces Published on disputed cells. Only three labels were offered; unadjudicated cells are assumed correct. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 14. Composite-reference paired differences by adjudicator

Adjudicator	Comparison	Domain	Weighted κ	AC2	Exact agreement
Adjudicator 1	ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5)	D1–D5	0.21 (0.14–0.29)	0.07 (0.04–0.09)	8.6 (5.7–11.6)
Adjudicator 1	ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5)	D1	0.03 (−0.09 to 0.16)	0.01 (−0.02 to 0.04)	0.0 (−3.9 to 3.9)
Adjudicator 1	ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5)	D2	0.35 (0.11–0.62)	0.09 (0.04–0.15)	12.7 (4.9–20.6)
Adjudicator 1	ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5)	D3	0.12 (−0.02 to 0.26)	0.05 (−0.02 to 0.12)	4.9 (−2.0 to 11.8)
Adjudicator 1	ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5)	D4	0.24 (0.02–0.46)	0.04 (0.01–0.07)	5.9 (1.0–11.8)
Adjudicator 1	ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5)	D5	0.36 (0.22–0.50)	0.17 (0.10–0.23)	19.6 (11.8–27.5)
Adjudicator 2	ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5)	D1–D5	0.24 (0.17–0.32)	0.08 (0.05–0.10)	9.6 (6.9–12.5)
Adjudicator 2	ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5)	D1	0.13 (0.03–0.25)	0.03 (0.01–0.07)	3.9 (1.0–7.8)
Adjudicator 2	ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5)	D2	0.21 (−0.03 to 0.47)	0.05 (−0.01 to 0.11)	5.9 (−2.9 to 13.7)
Adjudicator 2	ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5)	D3	0.24 (0.12–0.38)	0.12 (0.05–0.18)	12.7 (5.9–19.6)
Adjudicator 2	ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5)	D4	0.16 (−0.06 to 0.39)	0.02 (−0.01 to 0.06)	3.9 (−1.0 to 8.8)
Adjudicator 2	ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5)	D5	0.42 (0.28–0.54)	0.18 (0.12–0.24)	21.6 (14.7–28.4)

ARBITER minus Unassisted, coefficient units or percentage points; 510 paired cells per arm. Only offered labels could be chosen; unadjudicated cells are assumed correct. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 15. Preferred shares among adjudicated arm disagreements: sensitivities

Analysis	Measure	Estimate (95% CI)
Both adjudicators agreed	ARBITER	54/62 (87.1%; 79.0–94.3)
Both adjudicators agreed	Unassisted	7/62 (11.3%; 4.5–19.2)
Both adjudicators agreed	Neither	1/62 (1.6%; 0.0–4.8)
Adjudicator 1	ARBITER	60/80 (75.0%; 66.7–83.3)
Adjudicator 1	Unassisted	16/80 (20.0%; 11.7–28.9)
Adjudicator 1	Neither	4/80 (5.0%; 1.3–9.2)
Adjudicator 2	ARBITER	64/80 (80.0%; 72.3–87.2)
Adjudicator 2	Unassisted	15/80 (18.8%; 11.6–26.3)
Adjudicator 2	Neither	1/80 (1.2%; 0.0–3.8)
Counting split decisions against ARBITER	ARBITER	54/80 (67.5%; 58.7–76.3)
Counting split decisions against ARBITER	Unassisted	7/80 (8.8%; 3.5–15.0)
Counting split decisions against ARBITER	Neither	19/80 (23.8%; 16.0–31.3)

Fractions are adjudicated arm disagreements where the named rating was preferred / all adjudicated arm disagreements in that analysis. Consensus is primary and includes only cells with the same choice from both adjudicators; each single-adjudicator analysis includes cells where adjudicators split. Split cells are forced ARBITER losses in the split sensitivity; Neither includes these forced losses. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 16. Post hoc: preferred shares among adjudicated arm disagreements by strictness

Direction	Measure	Estimate (95% CI)
ARBITER stricter than Unassisted	ARBITER	44/45 (97.8%; 92.7–100.0)
ARBITER stricter than Unassisted	Unassisted	1/45 (2.2%; 0.0–7.3)
ARBITER stricter than Unassisted	Neither	0/45 (0.0%)
ARBITER more lenient than Unassisted	ARBITER	10/17 (58.8%; 35.7–80.0)
ARBITER more lenient than Unassisted	Unassisted	6/17 (35.3%; 14.3–57.9)
ARBITER more lenient than Unassisted	Neither	1/17 (5.9%; 0.0–18.2)

Post hoc analysis among adjudicated arm disagreements where both adjudicators agreed, stratified by ARBITER severity relative to Unassisted. Fractions are cells where the named rating was preferred / cells in that stratum. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 17. Independent and selected ratings: full agreement including AC2

Comparison	Weighted κ	AC2	Exact agreement
Adjudicator 1 vs Adjudicator 2 independent	0.59 (0.48–0.68)	0.69 (0.61–0.77)	114/154 (74.0%; 68.2–79.9)
Adjudicator 1 vs Adjudicator 2 preferred	0.69 (0.57–0.79)	0.83 (0.77–0.89)	129/154 (83.8%; 78.6–88.9)
Adjudicator 1 independent vs published label	−0.08 (−0.17 to 0.01)	−0.06 (−0.18 to 0.07)	34/154 (22.1%; 15.8–28.5)
Adjudicator 1 independent vs ARBITER (Sonnet 5.5)	0.52 (0.41–0.62)	0.66 (0.57–0.74)	107/154 (69.5%; 62.7–76.0)
Adjudicator 1 independent vs Unassisted (Sonnet 5.5)	0.22 (0.13–0.30)	0.28 (0.18–0.38)	69/154 (44.8%; 38.1–51.6)
Adjudicator 2 independent vs published label	−0.08 (−0.18 to 0.03)	0.14 (0.04–0.24)	42/154 (27.3%; 20.4–34.3)
Adjudicator 2 independent vs ARBITER (Sonnet 5.5)	0.67 (0.56–0.76)	0.79 (0.73–0.85)	120/154 (77.9%; 71.7–84.0)
Adjudicator 2 independent vs Unassisted (Sonnet 5.5)	0.31 (0.19–0.42)	0.45 (0.37–0.53)	76/154 (49.4%; 42.2–56.5)

154 disputed cells per comparison, selected because at least one Sonnet arm differed from the published review label. Independent ratings preceded label presentation; selected ratings chose among three offered labels. κ and AC2 are linearly weighted; raw agreement is n/N (x.x%; 95% CI a–b). This selected population cannot estimate full-benchmark agreement. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 18. Localisation population eligibility

Source	Ratings not chosen (cells)	Primary eligible (cells)	Sensitivity
ARBITER (Sonnet 5.5)	13	10	12
Unassisted (Sonnet 5.5)	60	47	57
Published	100	—	—

Primary requires both independent judgements equal to the adjudicators' agreed choice; sensitivity requires at least one. Published losses lack signalling answers. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 19. Error-localisation classes, both populations

Population	Class	ARBITER (Sonnet 5.5)	Unassisted (Sonnet 5.5)
Primary	Judgement departs from own answers	0	4
Primary	Single question that alone changed the rating	6	27
Primary	Several questions that alone changed the rating	1	2
Primary	Jointly decisive	2	11
Primary	Answers agree, judgement differs	1	2
Primary	Not attributable	0	1
Sensitivity	Judgement departs from own answers	0	4
Sensitivity	Single question that alone changed the rating	6	29
Sensitivity	Several questions that alone changed the rating	1	2
Sensitivity	Jointly decisive	4	18
Sensitivity	Answers agree, judgement differs	1	2
Sensitivity	Not attributable	0	2

Counts of losing cells. First applicable class is assigned; classes sum to eligible losses. No-discordance can include adjudicator disagreements, flagged separately. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 20. Questions that alone changed the rating: frequencies, both populations

Population	Domain	Question	Signalling question	ARBITER (Sonnet 5.5)	Unassisted (Sonnet 5.5)
Primary	D1	1.1	Random allocation sequence	0	0
Primary	D1	1.2	Concealment until enrolment and assignment	1	3
Primary	D1	1.3	Preintervention imbalance suggesting a randomisation problem	0	0
Primary	D2	2.1	Participants aware of assignment	0	0
Primary	D2	2.2	Carers or intervention providers aware of assignment	0	0
Primary	D2	2.3	Deviations arising from trial context	3	3
Primary	D2	2.4	Deviations affecting the outcome	0	0
Primary	D2	2.5	Deviations balanced across groups	0	0
Primary	D2	2.6	Appropriate analysis of assignment effect	0	3
Primary	D2	2.7	Substantial impact of analysing outside assigned groups	1	0
Primary	D3	3.1	Outcome data available for nearly everyone	0	1
Primary	D3	3.2	Evidence against bias from missing data	0	0
Primary	D3	3.3	Missingness potentially depends on true outcome	0	0
Primary	D3	3.4	Missingness likely depends on true outcome	0	0
Primary	D4	4.1	Inappropriate measurement method	0	0
Primary	D4	4.2	Measurement differs across groups	0	0
Primary	D4	4.3	Assessors aware of received intervention	0	0
Primary	D4	4.4	Assessment potentially influenced by awareness	0	0
Primary	D4	4.5	Assessment likely influenced by awareness	0	1
Primary	D5	5.1	Analysis plan finalised before unblinding	2	18
Primary	D5	5.2	Result selected from eligible outcome measurements	0	1
Primary	D5	5.3	Result selected from eligible analyses	1	1
Sensitivity	D1	1.1	Random allocation sequence	0	0
Sensitivity	D1	1.2	Concealment until enrolment and assignment	1	3
Sensitivity	D1	1.3	Preintervention imbalance suggesting a randomisation problem	0	0
Sensitivity	D2	2.1	Participants aware of assignment	0	0

Population	Domain	Question	Signalling question	ARBITER (Sonnet 5.5)	Unassisted (Sonnet 5.5)
Sensitivity	D2	2.2	Carers or intervention providers aware of assignment	0	0
Sensitivity	D2	2.3	Deviations arising from trial context	3	3
Sensitivity	D2	2.4	Deviations affecting the outcome	0	0
Sensitivity	D2	2.5	Deviations balanced across groups	0	0
Sensitivity	D2	2.6	Appropriate analysis of assignment effect	0	4
Sensitivity	D2	2.7	Substantial impact of analysing outside assigned groups	1	0
Sensitivity	D3	3.1	Outcome data available for nearly everyone	0	1
Sensitivity	D3	3.2	Evidence against bias from missing data	0	0
Sensitivity	D3	3.3	Missingness potentially depends on true outcome	0	0
Sensitivity	D3	3.4	Missingness likely depends on true outcome	0	0
Sensitivity	D4	4.1	Inappropriate measurement method	0	0
Sensitivity	D4	4.2	Measurement differs across groups	0	0
Sensitivity	D4	4.3	Assessors aware of received intervention	0	0
Sensitivity	D4	4.4	Assessment potentially influenced by awareness	0	0
Sensitivity	D4	4.5	Assessment likely influenced by awareness	0	1
Sensitivity	D5	5.1	Analysis plan finalised before unblinding	2	18
Sensitivity	D5	5.2	Result selected from eligible outcome measurements	0	2
Sensitivity	D5	5.3	Result selected from eligible analyses	1	1

Counts of questions that alone changed the rating; all 22 questions are retained. A cell may contribute several. Jointly decisive sets contribute none. Short question texts follow the 2019 guidance; D2 uses effect of assignment. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 21. Post hoc: supplied documents and answers in question 5.1 losses

Trial	Registry, protocol or SAP supplied	Unassisted mentions these documents	Unassisted answer	Adjudicators' answer
Allen et al., 2014	No	No	PY	NI
Amini et al., 2022	No	No	PY	NI
Apalla et al., 2013	No	No	PY	NI
Arezzo et al., 2008	Yes	No	PY	NI
ASCET	Yes	No	PY	NI
Attal et al., 2016	Yes	No	PY	NI
AWARD-1, 2014	Yes	No	PY	NI
Banovic 2024 (AVATAR)	Yes	No	PY	NI
Bente 2014	No	No	Y	NI
De Waal 2021	Yes	No	PY	NI
DURATION-8, 2016	Yes	No	PY	NI
Einarsson 2017	Yes	No	PY	NI
El-Khuffash 2021	No	No	PY	NI
GetGoal-Duo1, 2013	Yes	No	PY	NI
GETGOAL-F1, 2013	Yes	No	PY	NI
Levin 2011	Yes	No	PY	NI
Potsiurko 2022	Yes	No	PY	NI
TESTING	Yes	No	Y	N

Post hoc description of 18 primary Unassisted losses where question 5.1 alone changed the rating. Supplied means a bundle-inventoried registry, protocol or SAP; relevance/timing are not implied. De Waal's supplied registry is for another trial. Mentions include negated mentions in the Unassisted 5.1 justification. Both adjudicators agreed on each answer. Full justifications and inventories: question51-mechanism.csv. Y, yes; PY, probably yes; NI, no information. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 22. Unresolved independent answer

Trial	Domain	Adjudicator	Question	Recorded answer	Parsing issue
Chiba 2021	D3	Adjudicator 1	3.4	3.4=NI at 6 months; PY at 12 months	Ambiguous or unrecognised answer

The qualified answer is preserved without selecting a time point. This cell is outside both localisation populations because neither independent judgement equals the adjudicators' agreed choice. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 23. Post hoc: rules-only changes make three arm disagreements ties

Changed ratings (post hoc)

Trial	Domain	Original	Recomputed
An 2021	D1	Some concerns	High
Chiba 2021	D1	Some concerns	Low
Dian 2018	D1	Low	Some concerns
Gupta 2024	D1	Low	High
Sung 2020	D1	Low	Some concerns
Castoldi 2020	D2	Some concerns	High
Ibrahim 2017	D2	Some concerns	Low
Stopdapt	D2	Some concerns	High
Albert 2013	D3	Some concerns	High
Ashar 2022	D3	Some concerns	High
Levin 2016	D3	Some concerns	High
Bendix 1996	D4	Some concerns	Low
Boedt 2023	D4	Low	High
Fulop 2021	D4	Some concerns	Low
Abbasi 2012	D5	Some concerns	High
Aboelfadl 2022	D5	Some concerns	High
Attal 2021	D5	Some concerns	High
AWARD-CHN3 2023	D5	Low	Some concerns
Baillard 2018	D5	Low	Some concerns
Chiba 2021	D5	Some concerns	High
Dian 2018	D5	Some concerns	High
Envision	D5	Some concerns	High
Grant 2017	D5	Some concerns	High
Jindani 2016	D5	Low	Some concerns
Stopdapt	D5	Some concerns	High

Trial	Domain	Original	Recomputed
Sung 2020	D5	Some concerns	High

Post hoc. Three of 62 adjudicated arm disagreements changed to ARBITER's preferred rating; 59 were unchanged. Only violations are recomputed from Unassisted answers; invalid paths stay as reported. The original denominator is retained. Three new ties credit both arms, so preferred shares overlap. Weighted κ; exact agreement is n/N (%). CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 23 (continued). Post hoc: rules-only changes make three arm disagreements ties

Preferred shares on original adjudicated arm disagreements (post hoc)

Version	Measure	Estimate (95% CI)
Reported	ARBITER	54/62 (87.1%; 79.0–94.3)
Reported	Unassisted	7/62 (11.3%; 4.5–19.2)
Reported	Neither	1/62 (1.6%; 0.0–4.8)
Recomputed	ARBITER	54/62 (87.1%; 79.0–94.3)
Recomputed	Unassisted	10/62 (16.1%; 8.5–24.6)
Recomputed	Neither	1/62 (1.6%; 0.0–4.8)

Post hoc. Three of 62 adjudicated arm disagreements changed to ARBITER's preferred rating; 59 were unchanged. Only violations are recomputed from Unassisted answers; invalid paths stay as reported. The original denominator is retained. Three new ties credit both arms, so preferred shares overlap. Weighted κ; exact agreement is n/N (%). CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 23 (continued). Post hoc: rules-only changes make three arm disagreements ties

Agreement with published labels (post hoc)

Version	Domain	Measure	Estimate (95% CI)
Reported	D1–D5	Weighted κ	0.28 (0.21–0.35)
Reported	D1–D5	Exact agreement	395/510 (77.5%; 74.1–80.8)
Reported	D1	Weighted κ	0.39 (0.21–0.55)
Reported	D1	Exact agreement	82/102 (80.4%; 74.5–86.3)
Reported	D2	Weighted κ	0.06 (−0.10 to 0.21)
Reported	D2	Exact agreement	73/102 (71.6%; 64.7–78.4)
Reported	D3	Weighted κ	0.29 (0.13–0.44)
Reported	D3	Exact agreement	77/102 (75.5%; 68.6–82.4)
Reported	D4	Weighted κ	0.43 (0.22–0.61)
Reported	D4	Exact agreement	91/102 (89.2%; 83.3–94.1)
Reported	D5	Weighted κ	0.25 (0.07–0.43)
Reported	D5	Exact agreement	72/102 (70.6%; 63.7–78.4)
Recomputed	D1–D5	Weighted κ	0.24 (0.17–0.31)
Recomputed	D1–D5	Exact agreement	385/510 (75.5%; 72.0–78.8)
Recomputed	D1	Weighted κ	0.34 (0.17–0.50)
Recomputed	D1	Exact agreement	80/102 (78.4%; 71.6–85.3)
Recomputed	D2	Weighted κ	0.06 (−0.09 to 0.21)
Recomputed	D2	Exact agreement	73/102 (71.6%; 64.7–78.4)
Recomputed	D3	Weighted κ	0.26 (0.12–0.41)
Recomputed	D3	Exact agreement	75/102 (73.5%; 66.7–79.4)
Recomputed	D4	Weighted κ	0.34 (0.09–0.53)
Recomputed	D4	Exact agreement	90/102 (88.2%; 83.3–93.1)
Recomputed	D5	Weighted κ	0.17 (0.02–0.33)
Recomputed	D5	Exact agreement	67/102 (65.7%; 57.8–73.5)

Post hoc. Three of 62 adjudicated arm disagreements changed to ARBITER's preferred rating; 59 were unchanged. Only violations are recomputed from Unassisted answers; invalid paths stay as reported. The original denominator is retained. Three new ties credit both arms, so preferred shares overlap. Weighted κ; exact agreement is n/N (%). CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 23 (continued). Post hoc: rules-only changes make three arm disagreements ties

Agreement with the adjudicators' agreed choice (post hoc)

Version	Estimate (95% CI)
Reported	69/129 (53.5%; 46.0–60.7)
Recomputed	64/129 (49.6%; 41.7–57.4)

Post hoc. Three of 62 adjudicated arm disagreements changed to ARBITER's preferred rating; 59 were unchanged. Only violations are recomputed from Unassisted answers; invalid paths stay as reported. The original denominator is retained. Three new ties credit both arms, so preferred shares overlap. Weighted κ; exact agreement is n/N (%). CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 24. RoB 2 consistency and structural audit, all sources and domains

Domain	Measure	ARBITER (Sonnet 5.5)	Unassisted (Sonnet 5.5)	Published
Overall	Consistent with RoB 2	101/102 (99.0%)	102/102 (100.0%)	74/81 (91.4%)
Overall	Permitted escalation	1/102 (1.0%)	0/102 (0.0%)	6/81 (7.4%)
Overall	Rating contradicts its own answers	0/102 (0.0%)	0/102 (0.0%)	1/81 (1.2%)
D1–D5	Consistent with RoB 2	510/510 (100.0%)	444/509 (87.2%)	—
D1–D5	Proposed judgement matches algorithm	510/510 (100.0%)	—	—
D1–D5	Recorded override	0/510 (0.0%)	—	—
D1–D5	Rating contradicts its own answers	0/510 (0.0%)	26/509 (5.1%)	—
D1–D5	Required question skipped	0/510 (0.0%)	39/509 (7.7%)	—
D1–D5	Any rule break	0/510 (0.0%)	65/509 (12.8%)	—
D1	Consistent with RoB 2	102/102 (100.0%)	96/101 (95.0%)	—
D1	Proposed judgement matches algorithm	102/102 (100.0%)	—	—
D1	Recorded override	0/102 (0.0%)	—	—
D1	Rating contradicts its own answers	0/102 (0.0%)	5/101 (5.0%)	—
D1	Required question skipped	0/102 (0.0%)	0/101 (0.0%)	—
D1	Any rule break	0/102 (0.0%)	5/101 (5.0%)	—
D2	Consistent with RoB 2	102/102 (100.0%)	95/102 (93.1%)	—
D2	Proposed judgement matches algorithm	102/102 (100.0%)	—	—
D2	Recorded override	0/102 (0.0%)	—	—
D2	Rating contradicts its own answers	0/102 (0.0%)	3/102 (2.9%)	—
D2	Required question skipped	0/102 (0.0%)	4/102 (3.9%)	—
D2	Any rule break	0/102 (0.0%)	7/102 (6.9%)	—
D3	Consistent with RoB 2	102/102 (100.0%)	97/102 (95.1%)	—
D3	Proposed judgement matches algorithm	102/102 (100.0%)	—	—
D3	Recorded override	0/102 (0.0%)	—	—
D3	Rating contradicts its own answers	0/102 (0.0%)	3/102 (2.9%)	—
D3	Required question skipped	0/102 (0.0%)	2/102 (2.0%)	—
D3	Any rule break	0/102 (0.0%)	5/102 (4.9%)	—

Domain	Measure	ARBITER (Sonnet 5.5)	Unassisted (Sonnet 5.5)	Published
D4	Consistent with RoB 2	102/102 (100.0%)	67/102 (65.7%)	—
D4	Proposed judgement matches algorithm	102/102 (100.0%)	—	—
D4	Recorded override	0/102 (0.0%)	—	—
D4	Rating contradicts its own answers	0/102 (0.0%)	3/102 (2.9%)	—
D4	Required question skipped	0/102 (0.0%)	32/102 (31.4%)	—
D4	Any rule break	0/102 (0.0%)	35/102 (34.3%)	—
D5	Consistent with RoB 2	102/102 (100.0%)	89/102 (87.3%)	—
D5	Proposed judgement matches algorithm	102/102 (100.0%)	—	—
D5	Recorded override	0/102 (0.0%)	—	—
D5	Rating contradicts its own answers	0/102 (0.0%)	12/102 (11.8%)	—
D5	Required question skipped	0/102 (0.0%)	1/102 (1.0%)	—
D5	Any rule break	0/102 (0.0%)	13/102 (12.7%)	—
Audit	Answers citing ≥1 supporting passage or image	1538/1538 (100.0%)	—	—
Audit	Audit file passes the independent verifier	102/102 (100.0%)	—	—
Audit	All evidence bound	1421/1421 (100.0%)	—	—
Audit	Quoted text linked to its source	1407/1407 (100.0%)	—	—
Audit	Image regions linked to their page	14/14 (100.0%)	—	—
Audit	Other evidence records	0/1421 (0.0%)	—	—
Audit	Trials using visual evidence	8/102 (7.8%)	—	—

Values are n/N (%). Proposed judgements determine ARBITER domain consistency; adopted overrides are separate. Overall exact worst-domain agreement and permitted escalation are separate. Evidence coverage uses active signalling answers as the denominator and answers citing at least one evidence record as the numerator. Unassisted and Published evidence traces are unavailable. Binding and independent verification are structural checks, not clinical correctness. Visual transcriptions are model-written and not checked against images. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 25. Required question skipped sensitivity

Domain	Evaluable (n)	Original violations (n)	Invalid paths (n)	Additional violations (n)	Total violations (n)
Unassisted (Sonnet 5.5) — replacement: N/PN					
D1–D5	509	26	39	3	29
D1	101	5	0	0	5
D2	102	3	4	1	4
D3	102	3	2	2	5
D4	102	3	32	0	3
D5	102	12	1	0	12
Unassisted (Sonnet 5.5) — replacement: NI					
D1–D5	509	26	39	36	62
D1	101	5	0	0	5
D2	102	3	4	2	5
D3	102	3	2	2	5
D4	102	3	32	32	35
D5	102	12	1	0	12

Required invalid answers recursively resolve as N/PN then NI, including newly required questions. No off-path imputation; these are counts, not new assessments. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 26. Quotation syntax in active justifications

Source	Estimate (95% CI)
ARBITER (Sonnet 5.5)	34/1538 (2.2%; 1.1–3.5)
Unassisted (Sonnet 5.5)	0/1459 (0.0%)

Quoted active answers / all active answers. Paired double quotes enclose at least five words. Syntax is not verification and understates ARBITER bundle-based grounding. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 27. Unassisted justification content

Source	Answer-code-only / exported fields (%)
Unassisted (Sonnet 5.5)	1344/1459 (92.1%)

Codes are Y, PY, PN, N, NI or NA. This describes exported fields without inferring absent reasoning. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 28. Time, cost and token use per trial by model

Model	Metric	ARBITER	Unassisted	Paired mean difference (95% CI)
Sonnet 5.5	System runtime (min/trial)	7.6 [6.1–10.5]	0.5 [0.4–0.5]	11.0 (8.4–14.1)
Sonnet 5.5	API list-price equivalent (US$/trial)	3.78 [3.46–4.10]	0.09 [0.07–0.13]	3.68 (3.61–3.75)
Sonnet 5.5	Input tokens (count/trial)	96 [86–104]	6 [6–6]	90 (88–92)
Sonnet 5.5	Cache creation tokens (millions/trial)	0.49 [0.46–0.54]	0.02 [0.01–0.03]	0.48 (0.47–0.49)
Sonnet 5.5	Cache read tokens (millions/trial)	4.96 [4.48–5.59]	0.02 [0.02–0.02]	4.99 (4.87–5.12)
Sonnet 5.5	Output tokens (thousands/trial)	21.4 [18.9–24.3]	2.4 [1.9–2.9]	19.5 (18.8–20.1)
Sonnet 5.5	Thinking tokens (thousands/trial)	3.5 [2.6–4.2]	1.0 [0.7–1.3]	2.5 (2.3–2.7)
Sonnet 5.5	Agent steps/trial	60.0 [54.0–71.8]	4.0 [4.0–5.0]	58.6 (56.9–60.3)
Sonnet 5.5	Continue prompts/trial	0.0 [0.0–0.0]	0.0 [0.0–0.0]	0.0 (0.0–0.0)
Sonnet 5.5	Turns/trial	2.0 [2.0–2.0]	1.0 [1.0–1.0]	1.0 (1.0–1.0)

102 trials per arm; per-trial median [IQR] and paired mean ARBITER minus Unassisted difference (95% CI). US$ denotes API list-price equivalent; runs used a subscription. Thinking is included in output; eMethods give the reconstruction formula and rates. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 29. API cost reconstruction: estimates and bounds

Measure	ARBITER (Sonnet 5.5)	Unassisted (Sonnet 5.5)
Primary estimate: total (USD)	386.07	10.49
Primary estimate: median per trial (USD)	3.78	0.09
Lower reconstruction bound: total (USD)	386.07	10.49
Lower bound: median per trial (USD)	3.78	0.09
Upper reconstruction bound: total (USD)	386.07	10.49
Upper bound: median per trial (USD)	3.78	0.09
Harness diagnostic: total (USD)	1140.34	27.12
Harness diagnostic: median per trial (USD)	11.11	0.24
5-minute write sensitivity: total (USD)	365.88	10.49
5-minute write sensitivity: median per trial (USD)	3.58	0.09
5-minute sensitivity lower bound: total (USD)	365.88	10.49
5-minute sensitivity lower bound: median per trial (USD)	3.58	0.09
5-minute sensitivity upper bound: total (USD)	365.88	10.49
5-minute sensitivity upper bound: median per trial (USD)	3.58	0.09

USD. Reconstruction bounds are not sampling CIs. Five-minute side-call cache writes are a sensitivity. Harness figures are unusable: unrecognised model and unknown cost basis. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 30. Token reconciliation and compaction diagnostics

Measure	ARBITER (Sonnet 5.5)	Unassisted (Sonnet 5.5)
Main-thread request input tokens (n)	9,794	644
Main-thread request 1-hour cache write tokens (n)	50,698,919	1,910,531
Main-thread request 5-minute cache write tokens (n)	0	0
Main-thread request cache read tokens (n)	512,761,030	3,585,005
Terminal input tokens (n)	9,794	644
Terminal 1-hour cache write tokens (n)	50,695,878	1,910,531
Terminal 5-minute cache write tokens (n)	0	0
Terminal cache read tokens (n)	512,867,801	3,585,005
Terminal output tokens (n)	2,237,043	248,579
Side-call input tokens (n)	5,777,447	0
Side-call 1-hour cache write tokens (n)	13,464,429	0
Side-call 5-minute cache write tokens (n)	0	0
Side-call cache read tokens (n)	203,137,779	0
Side-call output tokens (n)	2,388,538	0
Terminal minus request input tokens (n)	0	0
Terminal minus request 1-hour cache write tokens (n)	−3,041	0
Terminal minus request 5-minute cache write tokens (n)	0	0
Terminal minus request cache read tokens (n)	106,771	0
Main-thread requests (n)	4,897	322
Requests with prompts >100,000 tokens (n)	3,272	0
Compaction events (n)	322	0
Turns with side calls (n)	122	0
Turns with request counts above terminal counts (n)	3	0
Side-call turns without compaction (n)	0	0

Token totals across trials. Terminal minus request is signed, including retry residuals. Side calls are total model usage minus terminal usage; output already includes thinking. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 31. Exploratory signalling-answer agreement

Adjudicator 1

SQ	Signalling question	ARBITER (Sonnet 5.5)	Unassisted (Sonnet 5.5)
1.1	Random allocation sequence	20/22 (90.9%; 78.3–100.0)	20/21 (95.2%; 85.7–100.0)
1.2	Concealment until enrolment and assignment	19/22 (86.4%; 72.0–100.0)	16/21 (76.2%; 57.9–93.8)
1.3	Preintervention imbalance suggesting a randomisation problem	12/22 (54.5%; 36.4–70.8)	9/21 (42.9%; 23.5–60.9)
2.1	Participants aware of assignment	34/40 (85.0%; 74.4–94.6)	34/40 (85.0%; 74.4–94.6)
2.2	Carers or intervention providers aware of assignment	37/40 (92.5%; 84.2–100.0)	36/40 (90.0%; 81.0–97.6)
2.3	Deviations arising from trial context	27/40 (67.5%; 53.1–80.6)	27/40 (67.5%; 53.8–81.0)
2.4	Deviations affecting the outcome	39/40 (97.5%; 92.5–100.0)	38/40 (95.0%; 87.8–100.0)
2.5	Deviations balanced across groups	39/40 (97.5%; 92.5–100.0)	39/40 (97.5%; 92.5–100.0)
2.6	Appropriate analysis of assignment effect	31/40 (77.5%; 65.7–88.6)	23/40 (57.5%; 44.7–70.6)
2.7	Substantial impact of analysing outside assigned groups	26/40 (65.0%; 51.4–77.5)	17/40 (42.5%; 30.0–55.6)
3.1	Outcome data available for nearly everyone	28/32 (87.5%; 75.7–97.0)	19/31 (61.3%; 45.2–77.4)
3.2	Evidence against bias from missing data	29/32 (90.6%; 80.0–100.0)	19/32 (59.4%; 43.2–75.0)
3.3	Missingness potentially depends on true outcome	28/32 (87.5%; 75.7–97.0)	14/31 (45.2%; 29.2–61.5)
3.4	Missingness likely depends on true outcome	16/32 (50.0%; 34.5–65.8)	11/31 (35.5%; 20.7–51.5)
4.1	Inappropriate measurement method	15/15 (100.0%)	15/15 (100.0%)
4.2	Measurement differs across groups	15/15 (100.0%)	15/15 (100.0%)
4.3	Assessors aware of received intervention	12/15 (80.0%; 58.8–100.0)	13/15 (86.7%; 69.2–100.0)
4.4	Assessment potentially influenced by awareness	13/15 (86.7%; 68.8–100.0)	8/15 (53.3%; 31.2–75.0)
4.5	Assessment likely influenced by awareness	12/15 (80.0%; 58.3–100.0)	5/15 (33.3%; 11.1–56.2)
5.1	Analysis plan finalised before unblinding	37/45 (82.2%; 73.8–90.2)	19/45 (42.2%; 29.8–55.0)
5.2	Result selected from eligible outcome measurements	21/45 (46.7%; 34.1–58.3)	15/44 (34.1%; 22.5–45.2)
5.3	Result selected from eligible analyses	6/45 (13.3%; 4.8–22.5)	3/45 (6.7%; 0.0–13.6)

SQ, signalling question. Correct collapsed answers / eligible questions (95% CI) on disputed domains. Independent answers define reference; Y/PY, N/PN, NI/NA collapse. Missing predictions are wrong. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 31 (continued). Exploratory signalling-answer agreement

Adjudicator 2

SQ	Signalling question	ARBITER (Sonnet 5.5)	Unassisted (Sonnet 5.5)
1.1	Random allocation sequence	21/22 (95.5%; 85.7–100.0)	19/21 (90.5%; 78.3–100.0)
1.2	Concealment until enrolment and assignment	20/22 (90.9%; 78.9–100.0)	14/21 (66.7%; 47.4–85.7)
1.3	Preintervention imbalance suggesting a randomisation problem	22/22 (100.0%)	19/21 (90.5%; 78.3–100.0)
2.1	Participants aware of assignment	37/40 (92.5%; 84.4–100.0)	37/40 (92.5%; 84.4–100.0)
2.2	Carers or intervention providers aware of assignment	38/40 (95.0%; 88.2–100.0)	37/40 (92.5%; 84.4–100.0)
2.3	Deviations arising from trial context	31/40 (77.5%; 65.0–88.6)	27/40 (67.5%; 54.3–80.0)
2.4	Deviations affecting the outcome	39/40 (97.5%; 92.1–100.0)	36/40 (90.0%; 80.6–97.6)
2.5	Deviations balanced across groups	39/40 (97.5%; 92.1–100.0)	39/40 (97.5%; 92.1–100.0)
2.6	Appropriate analysis of assignment effect	32/40 (80.0%; 68.6–90.9)	26/40 (65.0%; 51.4–77.5)
2.7	Substantial impact of analysing outside assigned groups	31/40 (77.5%; 65.8–89.2)	26/40 (65.0%; 51.7–77.8)
3.1	Outcome data available for nearly everyone	30/32 (93.8%; 84.8–100.0)	21/31 (67.7%; 52.8–82.8)
3.2	Evidence against bias from missing data	30/32 (93.8%; 84.8–100.0)	20/32 (62.5%; 46.9–78.1)
3.3	Missingness potentially depends on true outcome	30/32 (93.8%; 84.8–100.0)	16/31 (51.6%; 36.0–67.9)
3.4	Missingness likely depends on true outcome	24/32 (75.0%; 60.6–88.0)	11/31 (35.5%; 19.4–52.8)
4.1	Inappropriate measurement method	15/15 (100.0%)	15/15 (100.0%)
4.2	Measurement differs across groups	15/15 (100.0%)	15/15 (100.0%)
4.3	Assessors aware of received intervention	12/15 (80.0%; 58.8–100.0)	13/15 (86.7%; 69.2–100.0)
4.4	Assessment potentially influenced by awareness	13/15 (86.7%; 68.8–100.0)	8/15 (53.3%; 31.2–75.0)
4.5	Assessment likely influenced by awareness	11/15 (73.3%; 50.0–92.9)	7/15 (46.7%; 25.0–69.2)
5.1	Analysis plan finalised before unblinding	33/45 (73.3%; 61.0–85.1)	16/45 (35.6%; 23.1–48.7)
5.2	Result selected from eligible outcome measurements	40/45 (88.9%; 80.0–96.1)	31/44 (70.5%; 57.1–82.9)
5.3	Result selected from eligible analyses	34/45 (75.6%; 64.1–86.0)	32/45 (71.1%; 59.1–82.6)

SQ, signalling question. Correct collapsed answers / eligible questions (95% CI) on disputed domains. Independent answers define reference; Y/PY, N/PN, NI/NA collapse. Missing predictions are wrong. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

eTable 32. Evidence and quotation locations for the prior-work comparison

Study	Quoted cell evidence and locations	Source
[ref: Marshall 2016]	Code rules — No; Enforcement — No; Agent — No: “Support Vector Machine” — Methods, Machine-Learning Approach: learned document/sentence classifiers, not official decision-rule execution or a tool-calling agent. Three classifiers use held-out trial reports; model internals differ. Registry/protocol — No: “Linking with this dataset may yield better performance.” — Discussion, selective reporting limitation: protocol/registry linkage is future work; current models use trial-publication text. Identical inputs — Yes; Fixed-model comparison — No: “test data” — Evaluation, Task 1: the held-out duplicate-assessment trial set is used across three classifier variants. This compares different classifiers, not the same model with/without an assessment workflow. Reference; Independent reference — Partly: “blinded expert panel” — Introduction and Evaluation: historical independent Cochrane assessments supply RoB labels; newly recruited blinded experts assess supporting-text relevance only. No newly adjudicated RoB reference is implied.	PMC4713900, Methods and Evaluation.
[ref: Eisele-Metzger 2025]	Rule checks — Partly: “checked the 100 overall judgements” — Methods, Data analysis: overall-rule check only; 2/100 violations; not signalling-answer/domain conformance. Independent reference — Partly: “We reviewed all ‘two-level discrepancies’” — Methods, Data analysis: unblinded author review against sources, not an independent panel or replacement reference.	PMC12527486, Methods.
[ref: Huang 2025]	Code rules — Partly: “using Excel based on the RoB2 decision algorithm” — Discussion, Principal Findings: secondary spreadsheet derivation; executable formulas/automation are not established. Rule checks — Partly: “investigated the discrepancies” — Methods, Statistical Analysis: direct versus algorithm-derived judgments compared; no violation count/rate. Independent reference — Yes: “initially conducted independent judgments” — Methods, Establishment of the Criterion Standard: three trained reviewers, then consensus. Registry/protocol — No: “excluding Multimedia Appendices and protocols” — Discussion, Principal Findings; Limitations additionally excludes registration documents. Protocol availability is metadata, not delivery of a protocol.	PMC12238788, Methods and Discussion.
[ref: Lai 2026]	Independent reference — Partly: “manually corrected based on research team consensus” — Methods, Establishment of the Reference Standard; Results sensitivity analysis: corrections to Toouli assessments, not a new blinded reference.	PMC13614218, Methods and Results.
[ref: Nyrhi 2026]	Registry/protocol — Yes: “PDF-versions of their protocols when available” — Methods, Prompt development and standardisation: available protocols supplied; DeepSeek received copied protocol text (eAppendix 1). Missing protocols for some trials do not turn documented delivery into Partly; the column does not require a protocol for every trial.	PMC13054287, Methods; eAppendix 1.
[ref: Gandhi 2026]	Agent — Partly: “web-browsing capabilities enabled” — Methods, Data collection and RoB assessment: capability enabled; actual external searches not logged/quantified. Independent reference — Yes: “masked to the OSRA’s assessments” — Same section: three reviewers independently re-rated all 50 trials; consensus/fourth reviewer resolved their differences.	PMC13334998, Methods.

Study	Quoted cell evidence and locations	Source
[ref: Choi 2026]	Code rules — Yes: “implemented in code” — Methods, RoB 2.0 Algorithm Implementation: authors recomputed domain judgments. Enforcement — Partly: “official flowchart aggregation logic” — Methods, Study Design and Data: ROBoto2 component only; general Gemini/Gems emit ratings. Rule checks — Yes: “conformance was 43%” — Results, Algorithm Conformance and Internal Consistency: 43%, 51%, 60% across configurations; skip logic also checked. Agent — Partly: “reference knowledge” — Methods, Study Design and Data: Gems retrieval and ROBoto2 pipeline, not demonstrated autonomous tool calls. Independent reference — Yes: “One experienced reviewer’s assessment served as the reference.” — Methods, Study Design and Data: a single expert, not a blinded panel.	Publisher full article, Methods and Results.
[ref: Hevia 2025]	Code rules — Yes; Enforcement — Partly: “implement the logic provided by the ROB2 flowcharts” — Section 4, Domain-level and overall judgments: integrated software aggregation; not proven rejection of invalid/incomplete answers, and Choi found displayed inconsistencies. Audit file — Partly: “retains the original LLM responses and rationales” — Section 4, Collecting user feedback: original and human-edited records retained; no independently verified source-integrity bundle. Agent — Partly: “retrieved evidence paragraphs as context” — Section 4, Answering signaling questions: retrieval pipeline, not autonomous agent tool selection. Independent reference — Yes: “245 reports” — Section 5, Annotation procedure: manually assessed expert subset used for every LLM evaluation.	arXiv:2511.03048 v1, sections 4–5.
[ref: Liang 2026]	Quotation checks — Partly: “a verbatim quotation drawn from the retrieved trial text” — Methods, Retrieval-Based Evidence Support and Hallucination Assessment: GPT-5 judge supplies quoted support verdicts; no deterministic occurrence/hash check established. Audit file — Partly: “returned structured JSON” — Same section: verdict, quote and rationale are reviewable; no complete independently verifiable source bundle established. Agent — Partly: “post hoc claim verification” — Discussion, Limitations: BM25 plus judge pipeline evaluates generated claims; no autonomous assessment agent established.	PMC13628037, Methods and Discussion.
[ref: Saadi 2026]	Agent — Not reported: “customised agent-based implementations” — Methods, LLM assessment workflow: structured prompting is established, but neither retrieval nor external tool calling is demonstrated. Independent reference — Partly: “two separate teams of human reviewers” — Methods, Study design and data source: concordant historical review ratings selected, not new adjudication.	PMC13528376, Methods.

Study	Quoted cell evidence and locations	Source
This study (ARBITER)	Code rules — Yes: “ARBITER's server proposes the algorithm judgement.” — ANALYSIS-PLAN, section 5.4, resolution dated 18 October 2026. Enforcement — Yes: “The assessing model may adopt a different judgement only through a reasoned override” — Same resolution: enforced proposal/recording path, not a claim that overrides are impossible. Rule checks — Yes: “zero domain rule breaks by design” — Results, rule consistency; Table 3, ARBITER D1–D5 rule-break row (0/510). Quotation checks — Yes: “every quotation is found in a stored source” — ANALYSIS-PLAN, section 7, Resolution; audit-summary.csv reports 1407/1407 text-span and 14/14 visual evidence bindings. Audit file — Yes: “containing `canonical.json` and `verification.json`” — ANALYSIS-PLAN, section 5.8, bundle definition; audit-summary.csv reports 102/102 bundle verification. Agent — Yes: “its MCP server.” — ARBITER run input inventory, Setup. Independent reference — Yes: “Blinded dual adjudication of the disagreement cells” — ANALYSIS-PLAN, section 7, Action: adjudicators first rated independently, then chose among blinded labels. Registry/protocol — Yes: “the 60 protocol and SAP PDFs” — Unassisted run input inventory, Inputs; ARBITER run input inventory, opening paragraph: registry document acquisition enabled. Availability varies by trial; Yes records delivery of the complete acquired ancillary set to both arms, not that every trial had a protocol.	This study, Methods, Results and Table 3.

One row per Table 4 study; reference callouts resolve in the bibliography. Quotations condense documented feature decisions and their locations. NR means not reported, not absent; definitions are in Table 4. Text-relevance review is not risk-of-bias adjudication. Quotation occurrence does not establish support. Recorded overrides were permitted; none occurred in 510 domains. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet’s linearly weighted agreement coefficient; κ, Cohen’s kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

Prompt templates and a filled example

ARBITER template

/arbiter-assess Assess risk of bias for {outcome} defined as {definition} in {trial}.

ARBITER example

/arbiter-assess Assess risk of bias for Adverse events graded by CTCAE v4.0; 'Grade 3 (severe AEs)' row of Table 3; patients with grade 3 events in Table 4 defined as Table 3 Grade 3 (severe AEs) number of AEs: 600 mg 6, 900 mg 5, 1,200 mg 9 (all subjects 20); Grade 4: 0; Grade 5 (death): 1 per arm. Table 4 total no. of patients with grade 3 events: 2, 4, 5. Abstract: 'Grade 3 adverse events occurred in only 2 patients in the 600-mg arm, 4 patients in the 900-mg arm, and 5 patients in the 1,200-mg arm.' in Aarnoutse 2017.

Unassisted template

Assess risk of bias for {outcome} defined as {definition} in {trial}.

Unassisted output-format instruction

When you have finished, give your answers in this format: a table with one row per signalling question (columns: domain, question number, answer as Y, PY, PN, N, NI or NA), and then a final line exactly like "FINAL: D1=<judgement>; D2=<judgement>; D3=<judgement>; D4=<judgement>; D5=<judgement>; Overall=<judgement>", where each judgement is Low, Some concerns or High.

Run provenance

Model claude-sonnet-5-5; effort medium; Claude Code 2.1.278; isolated configuration homes. ARBITER 1.0.0, public commit 2f549ae30f056ff441ea61a04b719968716f2219. ARBITER with Sonnet ran 12 October 2026 (12:12-19:17 EDT); other arms ran 14 October 2026. Harness approved the proposed result unchanged and replied Approved.; uncoached Continue. replies were permitted at most twice per phase. Each trial ran once in each arm; run-to-run variability is not estimated. Unassisted access was limited to Read, Glob and Grep, without an MCP server, skills or web tools. Supplied registry Markdown and acquired protocol/SAP documents preserved input parity. Trial PDFs are not redistributed; source DOIs and FILES.csv hashes identify them. Bootstrap seed 20261018, 10,000 replicates; R/renv/Quarto versions are recorded in the project lockfile and session-info.txt.